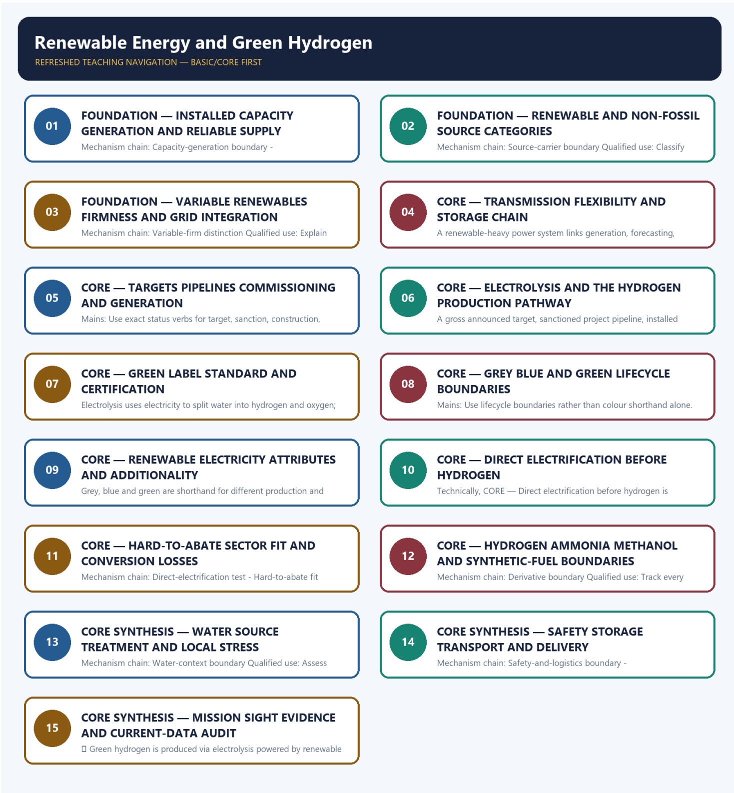
SOLVED PRACTICE WORKBOOK

Renewable Energy and Green Hydrogen - Solved Practice Workbook

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing



Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

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Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

Renewable Energy and Green Hydrogen - Solved Practice Workbook

Authoring-only generation: 2026-09-06. Uses the same source-bounded ecological distinctions and strict A-B-C-D rotation.

BASIC MCQS / REMEDIATION

Q1. Which statement correctly identifies Capacity-generation boundary?

A. Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply. B. Solar, wind, hydro, biomass and geothermal are renewable sources, while non-fossil is a wider category that can include nuclear power; the two labels are not interchangeable. C. Renewable electricity is an energy source output, while hydrogen is an energy carrier produced using an input pathway; hydrogen is not a primary renewable resource by itself. D. Solar and wind output varies with resource availability, whereas dispatchable or firm supply can be scheduled or supported to meet demand; nameplate capacity alone does not establish firmness.

Answer: A. Explanation: Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q2. Which option preserves the ecological boundary of Capacity-generation boundary?

A. Renewable electricity is an energy source output, while hydrogen is an energy carrier produced using an input pathway; hydrogen is not a primary renewable resource by itself. B. Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply. C. Solar and wind output varies with resource availability, whereas dispatchable or firm supply can be scheduled or supported to meet demand; nameplate capacity alone does not establish firmness. D. A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints.

Answer: B. Explanation: Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q3. Which statement uses Capacity-generation boundary without changing its scale, parameter or status?

A. Solar and wind output varies with resource availability, whereas dispatchable or firm supply can be scheduled or supported to meet demand; nameplate capacity alone does not establish firmness. B. A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints. C. Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply. D. Battery and pumped-storage systems shift energy across time and provide grid services, but storage consumes previously generated energy and is not an independent primary energy source.

Answer: C. Explanation: Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q4. Which option avoids the standard UPSC close-option trap about Capacity-generation boundary?

A. A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints. B. Battery and pumped-storage systems shift energy across time and provide grid services, but storage consumes previously generated energy and is not an independent primary energy source. C. A gross announced target, sanctioned project pipeline, installed capacity, commissioned capacity and achieved generation are separate statuses that require separate dated evidence. D. Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply.

Answer: D. Explanation: Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q5. Which statement correctly identifies Renewable-non-fossil boundary?

A. Solar, wind, hydro, biomass and geothermal are renewable sources, while non-fossil is a wider category that can include nuclear power; the two labels are not interchangeable. B. Renewable electricity is an energy source output, while hydrogen is an energy carrier produced using an input pathway; hydrogen is not a primary renewable resource by itself. C. Solar and wind output varies with resource availability, whereas dispatchable or firm supply can be scheduled or supported to meet demand; nameplate capacity alone does not establish firmness. D. A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints.

Answer: A. Explanation: Solar, wind, hydro, biomass and geothermal are renewable sources, while non-fossil is a wider category that can include nuclear power; the two labels are not interchangeable. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q6. Which option preserves the ecological boundary of Renewable-non-fossil boundary?

A. Solar and wind output varies with resource availability, whereas dispatchable or firm supply can be scheduled or supported to meet demand; nameplate capacity alone does not establish firmness. B. Solar, wind, hydro, biomass and geothermal are renewable sources, while non-fossil is a wider category that can include nuclear power; the two labels are not interchangeable. C. A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints. D. Battery and pumped-storage systems shift energy across time and provide grid services, but storage consumes previously generated energy and is not an independent primary energy source.

Answer: B. Explanation: Solar, wind, hydro, biomass and geothermal are renewable sources, while non-fossil is a wider category that can include nuclear power; the two labels are not interchangeable. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q7. Which statement uses Renewable-non-fossil boundary without changing its scale, parameter or status?

A. A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints. B. Battery and pumped-storage systems shift energy across time and provide grid services, but storage consumes previously generated energy and is not an independent primary energy source. C. Solar, wind, hydro, biomass and geothermal are renewable sources, while non-fossil is a wider category that can include nuclear power; the two labels are not interchangeable. D. A gross announced target, sanctioned project pipeline, installed capacity, commissioned capacity and achieved generation are separate statuses that require separate dated evidence.

Answer: C. Explanation: Solar, wind, hydro, biomass and geothermal are renewable sources, while non-fossil is a wider category that can include nuclear power; the two labels are not interchangeable. The other options

belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q8. Which option avoids the standard UPSC close-option trap about Renewable-non-fossil boundary?

A. Battery and pumped-storage systems shift energy across time and provide grid services, but storage consumes previously generated energy and is not an independent primary energy source. B. A gross announced target, sanctioned project pipeline, installed capacity, commissioned capacity and achieved generation are separate statuses that require separate dated evidence. C. Announced, approved, allocated, awarded, under construction, commissioned and producing describe different project stages; one stage must never be substituted for another. D. Solar, wind, hydro, biomass and geothermal are renewable sources, while non-fossil is a wider category that can include nuclear power; the two labels are not interchangeable.

Answer: D. Explanation: Solar, wind, hydro, biomass and geothermal are renewable sources, while non-fossil is a wider category that can include nuclear power; the two labels are not interchangeable. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q9. Which statement correctly identifies Source-carrier boundary?

A. Renewable electricity is an energy source output, while hydrogen is an energy carrier produced using an input pathway; hydrogen is not a primary renewable resource by itself. B. Solar and wind output varies with resource availability, whereas dispatchable or firm supply can be scheduled or supported to meet demand; nameplate capacity alone does not establish firmness. C. A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints. D. Battery and pumped-storage systems shift energy across time and provide grid services, but storage consumes previously generated energy and is not an independent primary energy source.

Answer: A. Explanation: Renewable electricity is an energy source output, while hydrogen is an energy carrier produced using an input pathway; hydrogen is not a primary renewable resource by itself. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q10. Which option preserves the ecological boundary of Source-carrier boundary?

A. A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints. B. Renewable electricity is an energy source output, while hydrogen is an energy carrier produced using an input pathway; hydrogen is not a primary renewable resource by itself. C. Battery and pumped-storage systems shift energy across time and provide grid services, but storage consumes previously generated energy and is not an independent primary energy source. D. A gross announced target, sanctioned project pipeline, installed capacity, commissioned capacity and achieved generation are separate statuses that require separate dated evidence.

Answer: B. Explanation: Renewable electricity is an energy source output, while hydrogen is an energy carrier produced using an input pathway; hydrogen is not a primary renewable resource by itself. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q11. Which statement uses Source-carrier boundary without changing its scale, parameter or status?

A. Battery and pumped-storage systems shift energy across time and provide grid services, but storage consumes previously generated energy and is not an independent primary energy source. B. A gross announced target, sanctioned project pipeline, installed capacity, commissioned capacity and achieved generation are separate statuses that require separate dated evidence. C. Renewable electricity is an energy source output, while hydrogen is an energy carrier produced using an input pathway; hydrogen is not a primary renewable resource by itself. D. Announced, approved, allocated, awarded, under construction, commissioned and producing describe different project stages; one stage must never be substituted for another.

Answer: C. Explanation: Renewable electricity is an energy source output, while hydrogen is an energy carrier produced using an input pathway; hydrogen is not a primary renewable resource by itself. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q12. Which option avoids the standard UPSC close-option trap about Source-carrier boundary?

A. A gross announced target, sanctioned project pipeline, installed capacity, commissioned capacity and achieved generation are separate statuses that require separate dated evidence. B. Announced, approved, allocated, awarded, under construction, commissioned and producing describe different project stages; one stage must never be substituted for another. C. Electrolysis uses electricity to split water into hydrogen and oxygen; calling the product green depends on the electricity and accounting pathway, not on the chemical identity of hydrogen. D. Renewable electricity is an energy source output, while hydrogen is an energy carrier produced using an input pathway; hydrogen is not a primary renewable resource by itself.

Answer: D. Explanation: Renewable electricity is an energy source output, while hydrogen is an energy carrier produced using an input pathway; hydrogen is not a primary renewable resource by itself. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q13. Which statement correctly identifies Variable-firm distinction?

A. Solar and wind output varies with resource availability, whereas dispatchable or firm supply can be scheduled or supported to meet demand; nameplate capacity alone does not establish firmness. B. A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints. C. Battery and pumped-storage systems shift energy across time and provide grid services, but storage consumes previously generated energy and is not an independent primary energy source. D. A gross announced target, sanctioned project pipeline, installed capacity, commissioned capacity and achieved generation are separate statuses that require separate dated evidence.

Answer: A. Explanation: Solar and wind output varies with resource availability, whereas dispatchable or firm supply can be scheduled or supported to meet demand; nameplate capacity alone does not establish firmness. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q14. Which option preserves the ecological boundary of Variable-firm distinction?

A. Battery and pumped-storage systems shift energy across time and provide grid services, but storage consumes previously generated energy and is not an independent primary energy source. B. Solar and wind output varies with resource availability, whereas dispatchable or firm supply can be scheduled or supported to meet demand; nameplate capacity alone does not establish firmness. C. A gross announced target, sanctioned project pipeline, installed capacity, commissioned capacity and achieved generation are separate statuses that require separate dated evidence. D. Announced, approved, allocated, awarded, under construction, commissioned and producing describe different project stages; one stage must never be substituted for another.

Answer: B. Explanation: Solar and wind output varies with resource availability, whereas dispatchable or firm supply can be scheduled or supported to meet demand; nameplate capacity alone does not establish firmness. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q15. Which statement uses Variable-firm distinction without changing its scale, parameter or status?

A. A gross announced target, sanctioned project pipeline, installed capacity, commissioned capacity and achieved generation are separate statuses that require separate dated evidence. B. Announced, approved, allocated, awarded, under construction, commissioned and producing describe different project stages; one stage must never be substituted for another. C. Solar and wind output varies with resource availability, whereas dispatchable or firm supply can be scheduled or supported to meet demand; nameplate capacity alone does not establish firmness. D. Electrolysis uses electricity to split water into hydrogen and oxygen; calling the product green depends on the electricity and accounting pathway, not on the chemical identity of hydrogen.

Answer: C. Explanation: Solar and wind output varies with resource availability, whereas dispatchable or firm supply can be scheduled or supported to meet demand; nameplate capacity alone does not establish firmness. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q16. Which option avoids the standard UPSC close-option trap about Variable-firm distinction?

A. Announced, approved, allocated, awarded, under construction, commissioned and producing describe different project stages; one stage must never be substituted for another. B. Electrolysis uses electricity to split water into hydrogen and oxygen; calling the product green depends on the electricity and accounting pathway, not on the chemical identity of hydrogen. C. A production pathway, a government definition or standard, a certification method and a market label are distinct layers; a pathway description alone does not prove certified compliance. D. Solar and wind output varies with resource availability, whereas dispatchable or firm supply can be scheduled or supported to meet demand; nameplate capacity alone does not establish firmness.

Answer: D. Explanation: Solar and wind output varies with resource availability, whereas dispatchable or firm supply can be scheduled or supported to meet demand; nameplate capacity alone does not establish firmness. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q17. Which statement correctly identifies Integration chain?

A. A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints. B. Battery and pumped-storage systems shift energy across time and provide grid services, but storage consumes previously generated energy and is not an independent primary energy source. C. A gross announced target, sanctioned project pipeline, installed capacity, commissioned capacity and achieved generation are separate statuses that require separate dated evidence. D. Announced, approved, allocated, awarded, under construction, commissioned and producing describe different project stages; one stage must never be substituted for another.

Answer: A. Explanation: A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q18. Which option preserves the ecological boundary of Integration chain?

A. A gross announced target, sanctioned project pipeline, installed capacity, commissioned capacity and achieved generation are separate statuses that require separate dated evidence. B. A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints. C. Announced, approved, allocated, awarded, under construction, commissioned and producing describe different project stages; one stage must never be substituted for another. D. Electrolysis uses electricity to split water into hydrogen and oxygen; calling the product green depends on the electricity and accounting pathway, not on the chemical identity of hydrogen.

Answer: B. Explanation: A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints. The other options belong to different media, parameters, processes, scales, actors,

institutions, instruments or status categories.

Q19. Which statement uses Integration chain without changing its scale, parameter or status?

A. Announced, approved, allocated, awarded, under construction, commissioned and producing describe different project stages; one stage must never be substituted for another. B. Electrolysis uses electricity to split water into hydrogen and oxygen; calling the product green depends on the electricity and accounting pathway, not on the chemical identity of hydrogen. C. A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints. D. A production pathway, a government definition or standard, a certification method and a market label are distinct layers; a pathway description alone does not prove certified compliance.

Answer: C. Explanation: A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q20. Which option avoids the standard UPSC close-option trap about Integration chain?

A. Electrolysis uses electricity to split water into hydrogen and oxygen; calling the product green depends on the electricity and accounting pathway, not on the chemical identity of hydrogen. B. A production pathway, a government definition or standard, a certification method and a market label are distinct layers; a pathway description alone does not prove certified compliance. C. Grey, blue and green are shorthand for different production and emissions-management pathways; colour language does not replace a stated lifecycle boundary, capture performance or official standard. D. A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints.

Answer: D. Explanation: A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q21. Which statement correctly identifies Storage boundary?

A. Battery and pumped-storage systems shift energy across time and provide grid services, but storage consumes previously generated energy and is not an independent primary energy source. B. A gross announced target, sanctioned project pipeline, installed capacity, commissioned capacity and achieved generation are separate statuses that require separate dated evidence. C. Announced, approved, allocated, awarded, under construction, commissioned and producing describe different project stages; one stage must never be substituted for another. D. Electrolysis uses electricity to split water into hydrogen and oxygen; calling the product green depends on the electricity and accounting pathway, not on the chemical identity of hydrogen.

Answer: A. Explanation: Battery and pumped-storage systems shift energy across time and provide grid services, but storage consumes previously generated energy and is not an independent primary energy source. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q22. Which option preserves the ecological boundary of Storage boundary?

A. Announced, approved, allocated, awarded, under construction, commissioned and producing describe different project stages; one stage must never be substituted for another. B. Battery and pumped-storage systems shift energy across time and provide grid services, but storage consumes previously generated energy and is not an independent primary energy source. C. Electrolysis uses electricity to split water into hydrogen and oxygen; calling the product green depends on the electricity and accounting pathway, not on the chemical identity of hydrogen. D. A production pathway, a government definition or standard, a certification method and a market label are distinct layers; a pathway description alone does not prove certified compliance.

Answer: B. Explanation: Battery and pumped-storage systems shift energy across time and provide grid services, but storage consumes previously generated energy and is not an independent primary energy

source. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q23. Which statement uses Storage boundary without changing its scale, parameter or status?

A. Electrolysis uses electricity to split water into hydrogen and oxygen; calling the product green depends on the electricity and accounting pathway, not on the chemical identity of hydrogen. B. A production pathway, a government definition or standard, a certification method and a market label are distinct layers; a pathway description alone does not prove certified compliance. C. Battery and pumped-storage systems shift energy across time and provide grid services, but storage consumes previously generated energy and is not an independent primary energy source. D. Grey, blue and green are shorthand for different production and emissions-management pathways; colour language does not replace a stated lifecycle boundary, capture performance or official standard.

Answer: C. Explanation: Battery and pumped-storage systems shift energy across time and provide grid services, but storage consumes previously generated energy and is not an independent primary energy source. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q24. Which option avoids the standard UPSC close-option trap about Storage boundary?

A. A production pathway, a government definition or standard, a certification method and a market label are distinct layers; a pathway description alone does not prove certified compliance. B. Grey, blue and green are shorthand for different production and emissions-management pathways; colour language does not replace a stated lifecycle boundary, capture performance or official standard. C. Claims about renewable electricity used for hydrogen may depend on sourcing, temporal matching, geography and additionality rules in the applicable standard; those attributes require documentary evidence. D. Battery and pumped-storage systems shift energy across time and provide grid services, but storage consumes previously generated energy and is not an independent primary energy source.

Answer: D. Explanation: Battery and pumped-storage systems shift energy across time and provide grid services, but storage consumes previously generated energy and is not an independent primary energy source. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q25. Which statement correctly identifies Target-achievement boundary?

A. A gross announced target, sanctioned project pipeline, installed capacity, commissioned capacity and achieved generation are separate statuses that require separate dated evidence. B. Announced, approved, allocated, awarded, under construction, commissioned and producing describe different project stages; one stage must never be substituted for another. C. Electrolysis uses electricity to split water into hydrogen and oxygen; calling the product green depends on the electricity and accounting pathway, not on the chemical identity of hydrogen. D. A production pathway, a government definition or standard, a certification method and a market label are distinct layers; a pathway description alone does not prove certified compliance.

Answer: A. Explanation: A gross announced target, sanctioned project pipeline, installed capacity, commissioned capacity and achieved generation are separate statuses that require separate dated evidence. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q26. Which option preserves the ecological boundary of Target-achievement boundary?

A. Electrolysis uses electricity to split water into hydrogen and oxygen; calling the product green depends on the electricity and accounting pathway, not on the chemical identity of hydrogen. B. A gross announced target, sanctioned project pipeline, installed capacity, commissioned capacity and achieved generation are separate statuses that require separate dated evidence. C. A production pathway, a government definition or standard, a certification method and a market label are distinct layers; a pathway description alone does not prove certified compliance. D. Grey, blue and green are shorthand for different production and emissions-management pathways; colour language does not replace a stated lifecycle boundary, capture performance or official standard.

Answer: B. Explanation: A gross announced target, sanctioned project pipeline, installed capacity, commissioned capacity and achieved generation are separate statuses that require separate dated evidence.

The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q27. Which statement uses Target-achievement boundary without changing its scale, parameter or status?

A. A production pathway, a government definition or standard, a certification method and a market label are distinct layers; a pathway description alone does not prove certified compliance. B. Grey, blue and green are shorthand for different production and emissions-management pathways; colour language does not replace a stated lifecycle boundary, capture performance or official standard. C. A gross announced target, sanctioned project pipeline, installed capacity, commissioned capacity and achieved generation are separate statuses that require separate dated evidence. D. Claims about renewable electricity used for hydrogen may depend on sourcing, temporal matching, geography and additionality rules in the applicable standard; those attributes require documentary evidence.

Answer: C. Explanation: A gross announced target, sanctioned project pipeline, installed capacity, commissioned capacity and achieved generation are separate statuses that require separate dated evidence. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q28. Which option avoids the standard UPSC close-option trap about Target-achievement boundary?

A. Grey, blue and green are shorthand for different production and emissions-management pathways; colour language does not replace a stated lifecycle boundary, capture performance or official standard. B. Claims about renewable electricity used for hydrogen may depend on sourcing, temporal matching, geography and additionality rules in the applicable standard; those attributes require documentary evidence. C. Direct use of clean electricity is generally the first option where technically suitable, while hydrogen is considered where molecules, high-temperature heat, long-duration storage or weight constraints matter. D. A gross announced target, sanctioned project pipeline, installed capacity, commissioned capacity and achieved generation are separate statuses that require separate dated evidence.

Answer: D. Explanation: A gross announced target, sanctioned project pipeline, installed capacity, commissioned capacity and achieved generation are separate statuses that require separate dated evidence. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q29. Which statement correctly identifies Status-verb discipline?

A. Announced, approved, allocated, awarded, under construction, commissioned and producing describe different project stages; one stage must never be substituted for another. B. Electrolysis uses electricity to split water into hydrogen and oxygen; calling the product green depends on the electricity and accounting pathway, not on the chemical identity of hydrogen. C. A production pathway, a government definition or standard, a certification method and a market label are distinct layers; a pathway description alone does not prove certified compliance. D. Grey, blue and green are shorthand for different production and emissions-management pathways; colour language does not replace a stated lifecycle boundary, capture performance or official standard.

Answer: A. Explanation: Announced, approved, allocated, awarded, under construction, commissioned and producing describe different project stages; one stage must never be substituted for another. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q30. Which option preserves the ecological boundary of Status-verb discipline?

A. A production pathway, a government definition or standard, a certification method and a market label are distinct layers; a pathway description alone does not prove certified compliance. B. Announced, approved, allocated, awarded, under construction, commissioned and producing describe different project stages; one stage must never be substituted for another. C. Grey, blue and green are shorthand for different production and emissions-management pathways; colour language does not replace a stated lifecycle boundary, capture performance or official standard. D. Claims about renewable electricity used for hydrogen may depend on sourcing, temporal matching, geography and additionality rules in the applicable standard; those attributes require documentary evidence.

Answer: B. Explanation: Announced, approved, allocated, awarded, under construction, commissioned and producing describe different project stages; one stage must never be substituted for another. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q31. Which statement uses Status-verb discipline without changing its scale, parameter or status?

A. Grey, blue and green are shorthand for different production and emissions-management pathways; colour language does not replace a stated lifecycle boundary, capture performance or official standard. B. Claims about renewable electricity used for hydrogen may depend on sourcing, temporal matching, geography and additionality rules in the applicable standard; those attributes require documentary evidence. C. Announced, approved, allocated, awarded, under construction, commissioned and producing describe different project stages; one stage must never be substituted for another. D. Direct use of clean electricity is generally the first option where technically suitable, while hydrogen is considered where molecules, high-temperature heat, long-duration storage or weight constraints matter.

Answer: C. Explanation: Announced, approved, allocated, awarded, under construction, commissioned and producing describe different project stages; one stage must never be substituted for another. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q32. Which option avoids the standard UPSC close-option trap about Status-verb discipline?

A. Claims about renewable electricity used for hydrogen may depend on sourcing, temporal matching, geography and additionality rules in the applicable standard; those attributes require documentary evidence. B. Direct use of clean electricity is generally the first option where technically suitable, while hydrogen is considered where molecules, high-temperature heat, long-duration storage or weight constraints matter. C. Fertiliser, refining, selected steel processes, shipping fuels and some heavy transport can use hydrogen or derivatives, but sector suitability and conversion losses must be assessed rather than assumed. D. Announced, approved, allocated, awarded, under construction, commissioned and producing describe different project stages; one stage must never be substituted for another.

Answer: D. Explanation: Announced, approved, allocated, awarded, under construction, commissioned and producing describe different project stages; one stage must never be substituted for another. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q33. Which statement correctly identifies Electrolysis pathway?

A. Electrolysis uses electricity to split water into hydrogen and oxygen; calling the product green depends on the electricity and accounting pathway, not on the chemical identity of hydrogen. B. A production pathway, a government definition or standard, a certification method and a market label are distinct layers; a pathway description alone does not prove certified compliance. C. Grey, blue and green are shorthand for different production and emissions-management pathways; colour language does not replace a stated lifecycle boundary, capture performance or official standard. D. Claims about renewable electricity used for hydrogen may depend on sourcing, temporal matching, geography and additionality rules in the applicable standard; those attributes require documentary evidence.

Answer: A. Explanation: Electrolysis uses electricity to split water into hydrogen and oxygen; calling the product green depends on the electricity and accounting pathway, not on the chemical identity of hydrogen. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q34. Which option preserves the ecological boundary of Electrolysis pathway?

A. Grey, blue and green are shorthand for different production and emissions-management pathways; colour language does not replace a stated lifecycle boundary, capture performance or official standard. B. Electrolysis uses electricity to split water into hydrogen and oxygen; calling the product green depends on the electricity and accounting pathway, not on the chemical identity of hydrogen. C. Claims about renewable electricity used for hydrogen may depend on sourcing, temporal matching, geography and additionality rules in the applicable standard; those attributes require documentary evidence. D. Direct use of clean electricity is generally the first option where technically suitable, while hydrogen is considered where molecules, high-temperature heat, long-duration storage or weight constraints matter.

Answer: B. Explanation: Electrolysis uses electricity to split water into hydrogen and oxygen; calling the product green depends on the electricity and accounting pathway, not on the chemical identity of hydrogen. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q35. Which statement uses Electrolysis pathway without changing its scale, parameter or status?

A. Claims about renewable electricity used for hydrogen may depend on sourcing, temporal matching, geography and additionality rules in the applicable standard; those attributes require documentary evidence. B. Direct use of clean electricity is generally the first option where technically suitable, while hydrogen is considered where molecules, high-temperature heat, long-duration storage or weight constraints matter. C. Electrolysis uses electricity to split water into hydrogen and oxygen; calling the product green depends on the electricity and accounting pathway, not on the chemical identity of hydrogen. D. Fertiliser, refining, selected steel processes, shipping fuels and some heavy transport can use hydrogen or derivatives, but sector suitability and conversion losses must be assessed rather than assumed.

Answer: C. Explanation: Electrolysis uses electricity to split water into hydrogen and oxygen; calling the product green depends on the electricity and accounting pathway, not on the chemical identity of hydrogen. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q36. Which option avoids the standard UPSC close-option trap about Electrolysis pathway?

A. Direct use of clean electricity is generally the first option where technically suitable, while hydrogen is considered where molecules, high-temperature heat, long-duration storage or weight constraints matter. B. Fertiliser, refining, selected steel processes, shipping fuels and some heavy transport can use hydrogen or derivatives, but sector suitability and conversion losses must be assessed rather than assumed. C. Hydrogen, ammonia, methanol and synthetic fuels are related but different products with different conversion steps, handling requirements and end uses; a hydrogen target is not automatically a derivative-output target. D. Electrolysis uses electricity to split water into hydrogen and oxygen; calling the product green depends on the electricity and accounting pathway, not on the chemical identity of hydrogen.

Answer: D. Explanation: Electrolysis uses electricity to split water into hydrogen and oxygen; calling the product green depends on the electricity and accounting pathway, not on the chemical identity of hydrogen. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q37. Which statement correctly identifies Label-standard-certification boundary?

A. A production pathway, a government definition or standard, a certification method and a market label are distinct layers; a pathway description alone does not prove certified compliance. B. Grey, blue and green are shorthand for different production and emissions-management pathways; colour language does not replace a stated lifecycle boundary, capture performance or official standard. C. Claims about renewable electricity used for hydrogen may depend on sourcing, temporal matching, geography and additionality rules in the applicable standard; those attributes require documentary evidence. D. Direct use of clean electricity is generally the first option where technically suitable, while hydrogen is considered where molecules, high-temperature heat, long-duration storage or weight constraints matter.

Answer: A. Explanation: A production pathway, a government definition or standard, a certification method and a market label are distinct layers; a pathway description alone does not prove certified compliance. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q38. Which option preserves the ecological boundary of Label-standard-certification boundary?

A. Claims about renewable electricity used for hydrogen may depend on sourcing, temporal matching, geography and additionality rules in the applicable standard; those attributes require documentary evidence. B. A production pathway, a government definition or standard, a certification method and a market label are distinct layers; a pathway description alone does not prove certified compliance. C. Direct use of clean electricity is generally the first option where technically suitable, while hydrogen is considered where molecules, high-temperature heat, long-duration storage or weight constraints matter. D. Fertiliser, refining, selected steel processes, shipping fuels and some heavy transport can use hydrogen or derivatives, but sector suitability and conversion losses must be assessed rather than assumed.

Answer: B. Explanation: A production pathway, a government definition or standard, a certification method and a market label are distinct layers; a pathway description alone does not prove certified compliance. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q39. Which statement uses Label-standard-certification boundary without changing its scale, parameter or status?

A. Direct use of clean electricity is generally the first option where technically suitable, while hydrogen is considered where molecules, high-temperature heat, long-duration storage or weight constraints matter. B. Fertiliser, refining, selected steel processes, shipping fuels and some heavy transport can use hydrogen or derivatives, but sector suitability and conversion losses must be assessed rather than assumed. C. A production pathway, a government definition or standard, a certification method and a market label are distinct layers; a pathway description alone does not prove certified compliance. D. Hydrogen, ammonia, methanol and synthetic fuels are related but different products with different conversion steps, handling requirements and end uses; a hydrogen target is not automatically a derivative-output target.

Answer: C. Explanation: A production pathway, a government definition or standard, a certification method and a market label are distinct layers; a pathway description alone does not prove certified compliance. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q40. Which option avoids the standard UPSC close-option trap about Label-standard-certification boundary?

A. Fertiliser, refining, selected steel processes, shipping fuels and some heavy transport can use hydrogen or derivatives, but sector suitability and conversion losses must be assessed rather than assumed. B. Hydrogen, ammonia, methanol and synthetic fuels are related but different products with different conversion steps, handling requirements and end uses; a hydrogen target is not automatically a derivative-output target. C. Electrolysis requires water and upstream treatment, but project-level water stress depends on source, quality, location, competing demand and management; no universal water-impact value should be inferred. D. A production pathway, a government definition or standard, a certification method and a market label are distinct layers; a pathway description alone does not prove certified compliance.

Answer: D. Explanation: A production pathway, a government definition or standard, a certification method and a market label are distinct layers; a pathway description alone does not prove certified compliance. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q41. Which statement correctly identifies Hydrogen-colour boundary?

A. Grey, blue and green are shorthand for different production and emissions-management pathways; colour language does not replace a stated lifecycle boundary, capture performance or official standard. B. Claims about renewable electricity used for hydrogen may depend on sourcing, temporal matching, geography and additionality rules in the applicable standard; those attributes require documentary evidence. C. Direct use of clean electricity is generally the first option where technically suitable, while hydrogen is considered where molecules, high-temperature heat, long-duration storage or weight constraints matter. D. Fertiliser, refining, selected steel processes, shipping fuels and some heavy transport can use hydrogen or derivatives, but sector suitability and conversion losses must be assessed rather than assumed.

Answer: A. Explanation: Grey, blue and green are shorthand for different production and emissions-management pathways; colour language does not replace a stated lifecycle boundary, capture performance or official standard. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q42. Which option preserves the ecological boundary of Hydrogen-colour boundary?

A. Direct use of clean electricity is generally the first option where technically suitable, while hydrogen is considered where molecules, high-temperature heat, long-duration storage or weight constraints matter. B. Grey, blue and green are shorthand for different production and emissions-management pathways; colour language does not replace a stated lifecycle boundary, capture performance or official standard. C. Fertiliser, refining, selected steel processes, shipping fuels and some heavy transport can use hydrogen or derivatives, but sector suitability and conversion losses must be assessed rather than assumed. D. Hydrogen, ammonia, methanol and synthetic fuels are related but different products with different conversion steps, handling requirements and end uses; a hydrogen target is not automatically a derivative-output target.

Answer: B. Explanation: Grey, blue and green are shorthand for different production and emissions-management pathways; colour language does not replace a stated lifecycle boundary, capture performance or official standard. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q43. Which statement uses Hydrogen-colour boundary without changing its scale, parameter or status?

A. Fertiliser, refining, selected steel processes, shipping fuels and some heavy transport can use hydrogen or derivatives, but sector suitability and conversion losses must be assessed rather than assumed. B. Hydrogen, ammonia, methanol and synthetic fuels are related but different products with different conversion steps, handling requirements and end uses; a hydrogen target is not automatically a derivative-output target. C. Grey, blue and green are shorthand for different production and emissions-management pathways; colour language does not replace a stated lifecycle boundary, capture performance or official standard. D. Electrolysis requires water and upstream treatment, but project-level water stress depends on source, quality, location, competing demand and management; no universal water-impact value should be inferred.

Answer: C. Explanation: Grey, blue and green are shorthand for different production and emissions-management pathways; colour language does not replace a stated lifecycle boundary, capture performance or official standard. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q44. Which option avoids the standard UPSC close-option trap about Hydrogen-colour boundary?

A. Hydrogen, ammonia, methanol and synthetic fuels are related but different products with different conversion steps, handling requirements and end uses; a hydrogen target is not automatically a derivative-output target. B. Electrolysis requires water and upstream treatment, but project-level water stress depends on source, quality, location, competing demand and management; no universal water-impact value should be inferred. C. Hydrogen production, compression, liquefaction, storage, pipelines, ports and end use create distinct engineering and safety requirements; production capacity does not prove delivered usable fuel. D. Grey, blue and green are shorthand for different production and emissions-management pathways; colour language does not replace a stated lifecycle boundary, capture performance or official standard.

Answer: D. Explanation: Grey, blue and green are shorthand for different production and emissions-management pathways; colour language does not replace a stated lifecycle boundary, capture performance or official standard. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q45. Which statement correctly identifies Electricity-attribute boundary?

A. Claims about renewable electricity used for hydrogen may depend on sourcing, temporal matching, geography and additionality rules in the applicable standard; those attributes require documentary evidence. B. Direct use of clean electricity is generally the first option where technically suitable, while hydrogen is considered where molecules, high-temperature heat, long-duration storage or weight constraints matter. C. Fertiliser, refining, selected steel processes, shipping fuels and some heavy transport can use hydrogen or derivatives, but sector suitability and conversion losses must be assessed rather than assumed. D. Hydrogen, ammonia, methanol and synthetic fuels are related but different products with different conversion steps, handling requirements and end uses; a hydrogen target is not automatically a derivative-output target.

Answer: A. Explanation: Claims about renewable electricity used for hydrogen may depend on sourcing, temporal matching, geography and additionality rules in the applicable standard; those attributes require documentary evidence. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q46. Which option preserves the ecological boundary of Electricity-attribute boundary?

A. Fertiliser, refining, selected steel processes, shipping fuels and some heavy transport can use hydrogen or derivatives, but sector suitability and conversion losses must be assessed rather than assumed. B. Claims about renewable electricity used for hydrogen may depend on sourcing, temporal matching, geography and additionality rules in the applicable standard; those attributes require documentary evidence. C. Hydrogen, ammonia, methanol and synthetic fuels are related but different products with different conversion steps, handling requirements and end uses; a hydrogen target is not automatically a derivative-output target. D. Electrolysis requires water and upstream treatment, but project-level water stress depends on source, quality, location, competing demand and management; no universal water-impact value should be inferred.

Answer: B. Explanation: Claims about renewable electricity used for hydrogen may depend on sourcing, temporal matching, geography and additionality rules in the applicable standard; those attributes require documentary evidence. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q47. Which statement uses Electricity-attribute boundary without changing its scale, parameter or status?

A. Hydrogen, ammonia, methanol and synthetic fuels are related but different products with different conversion steps, handling requirements and end uses; a hydrogen target is not automatically a derivative-output target. B. Electrolysis requires water and upstream treatment, but project-level water stress depends on source, quality, location, competing demand and management; no universal water-impact value should be inferred. C. Claims about renewable electricity used for hydrogen may depend on sourcing, temporal matching, geography and additionality rules in the applicable standard; those attributes require documentary evidence. D. Hydrogen production, compression, liquefaction, storage, pipelines, ports and end use create distinct engineering and safety requirements; production capacity does not prove delivered usable fuel.

Answer: C. Explanation: Claims about renewable electricity used for hydrogen may depend on sourcing, temporal matching, geography and additionality rules in the applicable standard; those attributes require documentary evidence. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q48. Which option avoids the standard UPSC close-option trap about Electricity-attribute boundary?

A. Electrolysis requires water and upstream treatment, but project-level water stress depends on source, quality, location, competing demand and management; no universal water-impact value should be inferred. B. Hydrogen production, compression, liquefaction, storage, pipelines, ports and end use create distinct engineering and safety requirements; production capacity does not prove delivered usable fuel. C. A mission target expresses intended future scale, while allocation, manufacturing award, commissioning, output and emissions outcome are separate evidence stages. D. Claims about renewable electricity used for hydrogen may depend on sourcing, temporal matching, geography and additionality rules in the applicable standard; those attributes require documentary evidence.

Answer: D. Explanation: Claims about renewable electricity used for hydrogen may depend on sourcing, temporal matching, geography and additionality rules in the applicable standard; those attributes require documentary evidence. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q49. Which statement correctly identifies Direct-electrification test?

A. Direct use of clean electricity is generally the first option where technically suitable, while hydrogen is considered where molecules, high-temperature heat, long-duration storage or weight constraints matter. B. Fertiliser, refining, selected steel processes, shipping fuels and some heavy transport can use hydrogen or derivatives, but sector suitability and conversion losses must be assessed rather than assumed. C. Hydrogen, ammonia, methanol and synthetic fuels are related but different products with different conversion steps, handling requirements and end uses; a hydrogen target is not automatically a derivative-output target. D. Electrolysis requires water and upstream treatment, but project-level water stress depends on source, quality, location, competing demand and management; no universal water-impact value should be inferred.

Answer: A. Explanation: Direct use of clean electricity is generally the first option where technically suitable, while hydrogen is considered where molecules, high-temperature heat, long-duration storage or weight constraints matter. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q50. Which option preserves the ecological boundary of Direct-electrification test?

A. Hydrogen, ammonia, methanol and synthetic fuels are related but different products with different conversion steps, handling requirements and end uses; a hydrogen target is not automatically a derivative-output target. B. Direct use of clean electricity is generally the first option where technically suitable, while hydrogen is considered where molecules, high-temperature heat, long-duration storage or weight constraints matter. C. Electrolysis requires water and upstream treatment, but project-level water stress depends on source, quality, location, competing demand and management; no universal water-impact value should be inferred. D. Hydrogen production, compression, liquefaction, storage, pipelines, ports and end use create distinct engineering and safety requirements; production capacity does not prove delivered usable fuel.

Answer: B. Explanation: Direct use of clean electricity is generally the first option where technically suitable, while hydrogen is considered where molecules, high-temperature heat, long-duration storage or weight constraints matter. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q51. Which statement uses Direct-electrification test without changing its scale, parameter or status?

A. Electrolysis requires water and upstream treatment, but project-level water stress depends on source, quality, location, competing demand and management; no universal water-impact value should be inferred. B. Hydrogen production, compression, liquefaction, storage, pipelines, ports and end use create distinct engineering and safety requirements; production capacity does not prove delivered usable fuel. C. Direct use of clean electricity is generally the first option where technically suitable, while hydrogen is considered where molecules, high-temperature heat, long-duration storage or weight constraints matter. D. A mission target expresses intended future scale, while allocation, manufacturing award, commissioning, output and emissions outcome are separate evidence stages.

Answer: C. Explanation: Direct use of clean electricity is generally the first option where technically suitable, while hydrogen is considered where molecules, high-temperature heat, long-duration storage or weight constraints matter. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q52. Which option avoids the standard UPSC close-option trap about Direct-electrification test?

A. Hydrogen production, compression, liquefaction, storage, pipelines, ports and end use create distinct engineering and safety requirements; production capacity does not prove delivered usable fuel. B. A mission target expresses intended future scale, while allocation, manufacturing award, commissioning, output and emissions outcome are separate evidence stages. C. The SIGHT programme distinguishes support for electrolyser manufacturing from support for green-hydrogen production; manufacturing capability and hydrogen output are related but not identical. D. Direct use of clean electricity is generally the first option where technically suitable, while hydrogen is considered where molecules, high-temperature heat, long-duration storage or weight constraints matter.

Answer: D. Explanation: Direct use of clean electricity is generally the first option where technically suitable, while hydrogen is considered where molecules, high-temperature heat, long-duration storage or weight constraints matter. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q53. Which statement correctly identifies Hard-to-abate fit?

A. Fertiliser, refining, selected steel processes, shipping fuels and some heavy transport can use hydrogen or derivatives, but sector suitability and conversion losses must be assessed rather than assumed. B. Hydrogen, ammonia, methanol and synthetic fuels are related but different products with different conversion steps, handling requirements and end uses; a hydrogen target is not automatically a derivative-output target. C. Electrolysis requires water and upstream treatment, but project-level water stress depends on source, quality, location, competing demand and management; no universal water-impact value should be inferred. D. Hydrogen production, compression, liquefaction, storage, pipelines, ports and end use create distinct engineering and safety requirements; production capacity does not prove delivered usable fuel.

Answer: A. Explanation: Fertiliser, refining, selected steel processes, shipping fuels and some heavy transport can use hydrogen or derivatives, but sector suitability and conversion losses must be assessed rather than assumed. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q54. Which option preserves the ecological boundary of Hard-to-abate fit?

A. Electrolysis requires water and upstream treatment, but project-level water stress depends on source, quality, location, competing demand and management; no universal water-impact value should be inferred. B. Fertiliser, refining, selected steel processes, shipping fuels and some heavy transport can use hydrogen or derivatives, but sector suitability and conversion losses must be assessed rather than assumed. C. Hydrogen production, compression, liquefaction, storage, pipelines, ports and end use create distinct engineering and safety requirements; production capacity does not prove delivered usable fuel. D. A mission target expresses intended future scale, while allocation, manufacturing award, commissioning, output and emissions outcome are separate evidence stages.

Answer: B. Explanation: Fertiliser, refining, selected steel processes, shipping fuels and some heavy transport can use hydrogen or derivatives, but sector suitability and conversion losses must be assessed

rather than assumed. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q55. Which statement uses Hard-to-abate fit without changing its scale, parameter or status?

A. Hydrogen production, compression, liquefaction, storage, pipelines, ports and end use create distinct engineering and safety requirements; production capacity does not prove delivered usable fuel. B. A mission target expresses intended future scale, while allocation, manufacturing award, commissioning, output and emissions outcome are separate evidence stages. C. Fertiliser, refining, selected steel processes, shipping fuels and some heavy transport can use hydrogen or derivatives, but sector suitability and conversion losses must be assessed rather than assumed. D. The SIGHT programme distinguishes support for electrolyser manufacturing from support for green-hydrogen production; manufacturing capability and hydrogen output are related but not identical.

Answer: C. Explanation: Fertiliser, refining, selected steel processes, shipping fuels and some heavy transport can use hydrogen or derivatives, but sector suitability and conversion losses must be assessed rather than assumed. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q56. Which option avoids the standard UPSC close-option trap about Hard-to-abate fit?

A. A mission target expresses intended future scale, while allocation, manufacturing award, commissioning, output and emissions outcome are separate evidence stages. B. The SIGHT programme distinguishes support for electrolyser manufacturing from support for green-hydrogen production; manufacturing capability and hydrogen output are related but not identical. C. Installed capacity, generation, project cost, mission target, achieved output, electrolyser scale, hydrogen classification and certification status require a dated MNRE, power-system or applicable standards source. D. Fertiliser, refining, selected steel processes, shipping fuels and some heavy transport can use hydrogen or derivatives, but sector suitability and conversion losses must be assessed rather than assumed.

Answer: D. Explanation: Fertiliser, refining, selected steel processes, shipping fuels and some heavy transport can use hydrogen or derivatives, but sector suitability and conversion losses must be assessed rather than assumed. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q57. Which statement correctly identifies Derivative boundary?

A. Hydrogen, ammonia, methanol and synthetic fuels are related but different products with different conversion steps, handling requirements and end uses; a hydrogen target is not automatically a derivative-output target. B. Electrolysis requires water and upstream treatment, but project-level water stress depends on source, quality, location, competing demand and management; no universal water-impact value should be inferred. C. Hydrogen production, compression, liquefaction, storage, pipelines, ports and end use create distinct engineering and safety requirements; production capacity does not prove delivered usable fuel. D. A mission target expresses intended future scale, while allocation, manufacturing award, commissioning, output and emissions outcome are separate evidence stages.

Answer: A. Explanation: Hydrogen, ammonia, methanol and synthetic fuels are related but different products with different conversion steps, handling requirements and end uses; a hydrogen target is not automatically a derivative-output target. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q58. Which option preserves the ecological boundary of Derivative boundary?

A. Hydrogen production, compression, liquefaction, storage, pipelines, ports and end use create distinct engineering and safety requirements; production capacity does not prove delivered usable fuel. B. Hydrogen, ammonia, methanol and synthetic fuels are related but different products with different conversion steps, handling requirements and end uses; a hydrogen target is not automatically a derivative-output target. C. A mission target expresses intended future scale, while allocation, manufacturing award, commissioning, output and emissions outcome are separate evidence stages. D. The SIGHT programme distinguishes support for electrolyser manufacturing from support for green-hydrogen production; manufacturing capability and hydrogen output are related but not identical.

Answer: B. Explanation: Hydrogen, ammonia, methanol and synthetic fuels are related but different products with different conversion steps, handling requirements and end uses; a hydrogen target is not automatically a

derivative-output target. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q59. Which statement uses Derivative boundary without changing its scale, parameter or status?

A. A mission target expresses intended future scale, while allocation, manufacturing award, commissioning, output and emissions outcome are separate evidence stages. B. The SIGHT programme distinguishes support for electrolyser manufacturing from support for green-hydrogen production; manufacturing capability and hydrogen output are related but not identical. C. Hydrogen, ammonia, methanol and synthetic fuels are related but different products with different conversion steps, handling requirements and end uses; a hydrogen target is not automatically a derivative-output target. D. Installed capacity, generation, project cost, mission target, achieved output, electrolyser scale, hydrogen classification and certification status require a dated MNRE, power-system or applicable standards source.

Answer: C. Explanation: Hydrogen, ammonia, methanol and synthetic fuels are related but different products with different conversion steps, handling requirements and end uses; a hydrogen target is not automatically a derivative-output target. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q60. Which option avoids the standard UPSC close-option trap about Derivative boundary?

A. The SIGHT programme distinguishes support for electrolyser manufacturing from support for green-hydrogen production; manufacturing capability and hydrogen output are related but not identical. B. Installed capacity, generation, project cost, mission target, achieved output, electrolyser scale, hydrogen classification and certification status require a dated MNRE, power-system or applicable standards source. C. Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply. D. Hydrogen, ammonia, methanol and synthetic fuels are related but different products with different conversion steps, handling requirements and end uses; a hydrogen target is not automatically a derivative-output target.

Answer: D. Explanation: Hydrogen, ammonia, methanol and synthetic fuels are related but different products with different conversion steps, handling requirements and end uses; a hydrogen target is not automatically a derivative-output target. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q61. Which statement correctly identifies Water-context boundary?

A. Electrolysis requires water and upstream treatment, but project-level water stress depends on source, quality, location, competing demand and management; no universal water-impact value should be inferred. B. Hydrogen production, compression, liquefaction, storage, pipelines, ports and end use create distinct engineering and safety requirements; production capacity does not prove delivered usable fuel. C. A mission target expresses intended future scale, while allocation, manufacturing award, commissioning, output and emissions outcome are separate evidence stages. D. The SIGHT programme distinguishes support for electrolyser manufacturing from support for green-hydrogen production; manufacturing capability and hydrogen output are related but not identical.

Answer: A. Explanation: Electrolysis requires water and upstream treatment, but project-level water stress depends on source, quality, location, competing demand and management; no universal water-impact value should be inferred. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q62. Which option preserves the ecological boundary of Water-context boundary?

A. A mission target expresses intended future scale, while allocation, manufacturing award, commissioning, output and emissions outcome are separate evidence stages. B. Electrolysis requires water and upstream treatment, but project-level water stress depends on source, quality, location, competing demand and management; no universal water-impact value should be inferred. C. The SIGHT programme distinguishes support for electrolyser manufacturing from support for green-hydrogen production; manufacturing capability and hydrogen output are related but not identical. D. Installed capacity, generation, project cost, mission target, achieved output, electrolyser scale, hydrogen classification and certification status require a dated MNRE, power-system or applicable standards source.

Answer: B. Explanation: Electrolysis requires water and upstream treatment, but project-level water stress depends on source, quality, location, competing demand and management; no universal water-impact value should be inferred. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q63. Which statement uses Water-context boundary without changing its scale, parameter or status?

A. The SIGHT programme distinguishes support for electrolyser manufacturing from support for green-hydrogen production; manufacturing capability and hydrogen output are related but not identical. B. Installed capacity, generation, project cost, mission target, achieved output, electrolyser scale, hydrogen classification and certification status require a dated MNRE, power-system or applicable standards source. C. Electrolysis requires water and upstream treatment, but project-level water stress depends on source, quality, location, competing demand and management; no universal water-impact value should be inferred. D. Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply.

Answer: C. Explanation: Electrolysis requires water and upstream treatment, but project-level water stress depends on source, quality, location, competing demand and management; no universal water-impact value should be inferred. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q64. Which option avoids the standard UPSC close-option trap about Water-context boundary?

A. Installed capacity, generation, project cost, mission target, achieved output, electrolyser scale, hydrogen classification and certification status require a dated MNRE, power-system or applicable standards source. B. Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply. C. Solar, wind, hydro, biomass and geothermal are renewable sources, while non-fossil is a wider category that can include nuclear power; the two labels are not interchangeable. D. Electrolysis requires water and upstream treatment, but project-level water stress depends on source, quality, location, competing demand and management; no universal water-impact value should be inferred.

Answer: D. Explanation: Electrolysis requires water and upstream treatment, but project-level water stress depends on source, quality, location, competing demand and management; no universal water-impact value should be inferred. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q65. Which statement correctly identifies Safety-and-logistics boundary?

A. Hydrogen production, compression, liquefaction, storage, pipelines, ports and end use create distinct engineering and safety requirements; production capacity does not prove delivered usable fuel. B. A mission target expresses intended future scale, while allocation, manufacturing award, commissioning, output and emissions outcome are separate evidence stages. C. The SIGHT programme distinguishes support for electrolyser manufacturing from support for green-hydrogen production; manufacturing capability and hydrogen output are related but not identical. D. Installed capacity, generation, project cost, mission target, achieved output, electrolyser scale, hydrogen classification and certification status require a dated MNRE, power-system or applicable standards source.

Answer: A. Explanation: Hydrogen production, compression, liquefaction, storage, pipelines, ports and end use create distinct engineering and safety requirements; production capacity does not prove delivered usable fuel. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q66. Which option preserves the ecological boundary of Safety-and-logistics boundary?

A. The SIGHT programme distinguishes support for electrolyser manufacturing from support for green-hydrogen production; manufacturing capability and hydrogen output are related but not identical. B. Hydrogen production, compression, liquefaction, storage, pipelines, ports and end use create distinct engineering and safety requirements; production capacity does not prove delivered usable fuel. C. Installed capacity, generation, project cost, mission target, achieved output, electrolyser scale, hydrogen classification and certification status require a dated MNRE, power-system or applicable standards source. D. Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply.

Answer: B. Explanation: Hydrogen production, compression, liquefaction, storage, pipelines, ports and end use create distinct engineering and safety requirements; production capacity does not prove delivered usable fuel. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q67. Which statement uses Safety-and-logistics boundary without changing its scale, parameter or status?

A. Installed capacity, generation, project cost, mission target, achieved output, electrolyser scale, hydrogen classification and certification status require a dated MNRE, power-system or applicable standards source. B. Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply. C. Hydrogen production, compression, liquefaction, storage, pipelines, ports and end use create distinct engineering and safety requirements; production capacity does not prove delivered usable fuel. D. Solar, wind, hydro, biomass and geothermal are renewable sources, while non-fossil is a wider category that can include nuclear power; the two labels are not interchangeable.

Answer: C. Explanation: Hydrogen production, compression, liquefaction, storage, pipelines, ports and end use create distinct engineering and safety requirements; production capacity does not prove delivered usable fuel. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q68. Which option avoids the standard UPSC close-option trap about Safety-and-logistics boundary?

A. Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply. B. Solar, wind, hydro, biomass and geothermal are renewable sources, while non-fossil is a wider category that can include nuclear power; the two labels are not interchangeable. C. Renewable electricity is an energy source output, while hydrogen is an energy carrier produced using an input pathway; hydrogen is not a primary renewable resource by itself. D. Hydrogen production, compression, liquefaction, storage, pipelines, ports and end use create distinct engineering and safety requirements; production capacity does not prove delivered usable fuel.

Answer: D. Explanation: Hydrogen production, compression, liquefaction, storage, pipelines, ports and end use create distinct engineering and safety requirements; production capacity does not prove delivered usable fuel. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q69. Which statement correctly identifies Mission-target-outcome boundary?

A. A mission target expresses intended future scale, while allocation, manufacturing award, commissioning, output and emissions outcome are separate evidence stages. B. The SIGHT programme distinguishes support for electrolyser manufacturing from support for green-hydrogen production; manufacturing capability and hydrogen output are related but not identical. C. Installed capacity, generation, project cost, mission target, achieved output, electrolyser scale, hydrogen classification and certification status require a dated MNRE, power-system or applicable standards source. D. Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply.

Answer: A. Explanation: A mission target expresses intended future scale, while allocation, manufacturing award, commissioning, output and emissions outcome are separate evidence stages. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q70. Which option preserves the ecological boundary of Mission-target-outcome boundary?

A. Installed capacity, generation, project cost, mission target, achieved output, electrolyser scale, hydrogen classification and certification status require a dated MNRE, power-system or applicable standards source. B. A mission target expresses intended future scale, while allocation, manufacturing award, commissioning, output and emissions outcome are separate evidence stages. C. Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply. D. Solar, wind, hydro, biomass and geothermal are renewable sources, while non-fossil is a wider category that can include nuclear power; the two labels are not interchangeable.

Answer: B. Explanation: A mission target expresses intended future scale, while allocation, manufacturing award, commissioning, output and emissions outcome are separate evidence stages. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q71. Which statement uses Mission-target-outcome boundary without changing its scale, parameter or status?

A. Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply. B. Solar, wind, hydro, biomass and geothermal are renewable sources, while non-fossil is a wider category that can include nuclear power; the two labels are not interchangeable. C. A mission target expresses intended future scale, while allocation, manufacturing award, commissioning, output and emissions outcome are separate evidence stages. D. Renewable electricity is an energy source output, while hydrogen is an energy carrier produced using an input pathway; hydrogen is not a primary renewable resource by itself.

Answer: C. Explanation: A mission target expresses intended future scale, while allocation, manufacturing award, commissioning, output and emissions outcome are separate evidence stages. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q72. Which option avoids the standard UPSC close-option trap about Mission-target-outcome boundary?

A. Solar, wind, hydro, biomass and geothermal are renewable sources, while non-fossil is a wider category that can include nuclear power; the two labels are not interchangeable. B. Renewable electricity is an energy source output, while hydrogen is an energy carrier produced using an input pathway; hydrogen is not a primary renewable resource by itself. C. Solar and wind output varies with resource availability, whereas dispatchable or firm supply can be scheduled or supported to meet demand; nameplate capacity alone does not establish firmness. D. A mission target expresses intended future scale, while allocation, manufacturing award, commissioning, output and emissions outcome are separate evidence stages.

Answer: D. Explanation: A mission target expresses intended future scale, while allocation, manufacturing award, commissioning, output and emissions outcome are separate evidence stages. The other options

belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q73. Which statement correctly identifies SIGHT-dual-pillar boundary?

A. The SIGHT programme distinguishes support for electrolyser manufacturing from support for green-hydrogen production; manufacturing capability and hydrogen output are related but not identical. B. Installed capacity, generation, project cost, mission target, achieved output, electrolyser scale, hydrogen classification and certification status require a dated MNRE, power-system or applicable standards source. C. Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply. D. Solar, wind, hydro, biomass and geothermal are renewable sources, while non-fossil is a wider category that can include nuclear power; the two labels are not interchangeable.

Answer: A. Explanation: The SIGHT programme distinguishes support for electrolyser manufacturing from support for green-hydrogen production; manufacturing capability and hydrogen output are related but not identical. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q74. Which option preserves the ecological boundary of SIGHT-dual-pillar boundary?

A. Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply. B. The SIGHT programme distinguishes support for electrolyser manufacturing from support for green-hydrogen production; manufacturing capability and hydrogen output are related but not identical. C. Solar, wind, hydro, biomass and geothermal are renewable sources, while non-fossil is a wider category that can include nuclear power; the two labels are not interchangeable. D. Renewable electricity is an energy source output, while hydrogen is an energy carrier produced using an input pathway; hydrogen is not a primary renewable resource by itself.

Answer: B. Explanation: The SIGHT programme distinguishes support for electrolyser manufacturing from support for green-hydrogen production; manufacturing capability and hydrogen output are related but not identical. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q75. Which statement uses SIGHT-dual-pillar boundary without changing its scale, parameter or status?

A. Solar, wind, hydro, biomass and geothermal are renewable sources, while non-fossil is a wider category that can include nuclear power; the two labels are not interchangeable. B. Renewable electricity is an energy source output, while hydrogen is an energy carrier produced using an input pathway; hydrogen is not a primary renewable resource by itself. C. The SIGHT programme distinguishes support for electrolyser manufacturing from support for green-hydrogen production; manufacturing capability and hydrogen output are related but not identical. D. Solar and wind output varies with resource availability, whereas dispatchable or firm supply can be scheduled or supported to meet demand; nameplate capacity alone does not establish firmness.

Answer: C. Explanation: The SIGHT programme distinguishes support for electrolyser manufacturing from support for green-hydrogen production; manufacturing capability and hydrogen output are related but not identical. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q76. Which option avoids the standard UPSC close-option trap about SIGHT-dual-pillar boundary?

A. Renewable electricity is an energy source output, while hydrogen is an energy carrier produced using an input pathway; hydrogen is not a primary renewable resource by itself. B. Solar and wind output varies with resource availability, whereas dispatchable or firm supply can be scheduled or supported to meet demand; nameplate capacity alone does not establish firmness. C. A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints. D. The SIGHT programme distinguishes support for electrolyser manufacturing from support for green-hydrogen production; manufacturing capability and hydrogen output are related but not identical.

Answer: D. Explanation: The SIGHT programme distinguishes support for electrolyser manufacturing from support for green-hydrogen production; manufacturing capability and hydrogen output are related but not identical. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q77. Which statement correctly identifies Current evidence boundary?

A. Installed capacity, generation, project cost, mission target, achieved output, electrolyser scale, hydrogen classification and certification status require a dated MNRE, power-system or applicable standards source. B. Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply. C. Solar, wind, hydro, biomass and geothermal are renewable sources, while non-fossil is a wider category that can include nuclear power; the two labels are not interchangeable. D. Renewable electricity is an energy source output, while hydrogen is an energy carrier produced using an input pathway; hydrogen is not a primary renewable resource by itself.

Answer: A. Explanation: Installed capacity, generation, project cost, mission target, achieved output, electrolyser scale, hydrogen classification and certification status require a dated MNRE, power-system or applicable standards source. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q78. Which option preserves the ecological boundary of Current evidence boundary?

A. Solar, wind, hydro, biomass and geothermal are renewable sources, while non-fossil is a wider category that can include nuclear power; the two labels are not interchangeable. B. Installed capacity, generation, project cost, mission target, achieved output, electrolyser scale, hydrogen classification and certification status require a dated MNRE, power-system or applicable standards source. C. Renewable electricity is an energy source output, while hydrogen is an energy carrier produced using an input pathway; hydrogen is not a primary renewable resource by itself. D. Solar and wind output varies with resource availability, whereas dispatchable or firm supply can be scheduled or supported to meet demand; nameplate capacity alone does not establish firmness.

Answer: B. Explanation: Installed capacity, generation, project cost, mission target, achieved output, electrolyser scale, hydrogen classification and certification status require a dated MNRE, power-system or applicable standards source. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q79. Which statement uses Current evidence boundary without changing its scale, parameter or status?

A. Renewable electricity is an energy source output, while hydrogen is an energy carrier produced using an input pathway; hydrogen is not a primary renewable resource by itself. B. Solar and wind output varies with resource availability, whereas dispatchable or firm supply can be scheduled or supported to meet demand; nameplate capacity alone does not establish firmness. C. Installed capacity, generation, project cost, mission target, achieved output, electrolyser scale, hydrogen classification and certification status require a dated MNRE, power-system or applicable standards source. D. A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints.

Answer: C. Explanation: Installed capacity, generation, project cost, mission target, achieved output, electrolyser scale, hydrogen classification and certification status require a dated MNRE, power-system or applicable standards source. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q80. Which option avoids the standard UPSC close-option trap about Current evidence boundary?

A. Solar and wind output varies with resource availability, whereas dispatchable or firm supply can be scheduled or supported to meet demand; nameplate capacity alone does not establish firmness. B. A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints. C. Battery and pumped-storage systems shift energy across time and provide grid services, but storage consumes previously generated energy and is not an independent primary energy source. D. Installed capacity, generation, project cost, mission target, achieved output, electrolyser scale, hydrogen classification and certification status require a dated MNRE, power-system or applicable standards source.

Answer: D. Explanation: Installed capacity, generation, project cost, mission target, achieved output, electrolyser scale, hydrogen classification and certification status require a dated MNRE, power-system or applicable standards source. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

PYQS AND ANSWER PRACTICE

AUDITED RENEWABLE, GRID, STORAGE, HYDROGEN AND ENERGY-INDEPENDENCE PYQ OWNERSHIP

Audited ledgers route renewable, solar, wind, grid, storage, bioenergy and hydrogen demands. No provisional objective key, installed-capacity value, generation figure, cost or achieved mission outcome is inferred.

OWNER PYQ LEDGER EXTRACTS

9. PYQ application

- FACT: **2025 GS-III direct PYQ (150 words):** "How can India achieve energy independence

through clean technology by 2047? How can biotechnology play a crucial role in this endeavour?" ANALYSIS: Two distinct demands. The **clean-technology** half is owned by this file (renewables, storage, nuclear/SHANTI Act, green hydrogen, grid and critical minerals); the **biotechnology** half needs bio-energy specifics - the National Bio-Energy Programme, compressed bio-gas, ethanol blending, enzymatic/algal routes to biofuels, and biomass cogeneration (9.82 GW grid-connected as of October 2025).

- FACT: **2025 GS-III cross-route (150 words):** the ITER/fusion question is a Science-and-

Technology demand, but this file supplies the correct framing for why fusion sits **outside** the 2030-2047 energy-transition arithmetic - it is a post-2050 option, not a Panchamrit instrument.

- ANALYSIS: Recurring Prelims pattern: distinguish green, grey and blue hydrogen by production

process and emissions profile, and identify the National Green Hydrogen Mission's key targets and the SIGHT Programme's function.

- ANALYSIS: Mains linkage: the hard-to-abate-sector rationale is used to argue for green hydrogen as

a complement to, not substitute for, direct electrification and renewable-capacity expansion.

2026 PYQ Integration

> **Status:** 2026 question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: `_PYQ-ROUTING-PRELIMS-2026.md`. **Answer-key rule:** The 2026 Prelims and CSAT Set-A keys held locally are **provisional**; no option or answer is recorded or inferred in this integration.

- **Year represented:** 2026
- **Paper(s):** Prelims GS-I
- **Routed question demands:** 1

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2026	Prelims GS-I	45	Green hydrogen production pathways and India's National Green Hydrogen Mission	Objective question; provisional 2026 Set-A key present locally, answer not inferred	Provisional 2026 Set-A key present locally (`Ans-■2026-■GS1-■Provisional`); key is provisional - no answer letter recorded or inferred here	Cover the named fact/concept and its likely statement-level distinctions.

What this owner must now support

- Green hydrogen production pathways and India's National Green Hydrogen Mission

> This block integrates the 2026 examinable demand and paper metadata. It is kept separate from the 2018-2023 and 2024-2025 blocks and does not convert a provisionally-keyed, answer-free objective question into a solved answer.

Recent PYQ Integration (2024-2025)

> **Status:** 2024-2025 question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: `_PYQ-ROUTING-MAINS-GS1-GS2-ESSAY-2024-2025.md`, `_PYQ-ROUTING-MAINS-GS3-GS4-2024-2025.md`, `_PYQ-ROUTING-PRELIMS-2024-2025.md`. **Answer-key rule:** The official 2024-2025 Prelims Set-A keys are present in the repository and CSAT Set-A keys are supplied; even so, no option or answer is recorded or inferred in this integration.

- **Years represented:** 2024, 2025
- **Paper(s):** GS-I, GS-III, Prelims GS-I
- **Routed question demands:** 6

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2024	Prelims GS-I	27	Distributed Energy Resources (battery storage, biomass, fuel cells, rooftop solar)	Objective question; official Set-A key available locally, answer not inferred	Key available locally (official Set-A answer key present); answer not recorded here	Cover the named fact/concept and its likely statement-level distinctions.
2024	Prelims GS-I	38	'Pumped-storage hydropower' (long-duration energy storage)	Objective question; official Set-A key available locally, answer not inferred	Key available locally (official Set-A answer key present); answer not recorded here	Cover the named fact/concept and its likely statement-level distinctions.
2024	Prelims GS-I	46	Feedstock for Sustainable Aviation Fuel	Objective question; official Set-A key available locally, answer not inferred	Key available locally (official Set-A answer key present); answer not recorded here	Cover the named fact/concept and its likely statement-level distinctions.
2025	GS-I	6	Ecological and economic benefits of solar energy generation in India	Explain · 10 marks · 150 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
2025	GS-III	6	Energy independence by 2047 through clean tech; role of biotechnology	Discuss · 10 marks · 150 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
2025	Prelims GS-I	70	PM Surya Ghar Muft Bijli Yojana - solar rooftop	Objective question; official Set-A key available locally, answer not inferred	Key available locally (official Set-A answer key present); answer not recorded here	Cover the named fact/concept and its likely statement-level distinctions.

What this owner must now support

- Distributed Energy Resources (battery storage, biomass, fuel cells, rooftop solar)

- 'Pumped-storage hydropower' (long-duration energy storage)
- Feedstock for Sustainable Aviation Fuel
- Ecological and economic benefits of solar energy generation in India
- Energy independence by 2047 through clean tech; role of biotechnology
- PM Surya Ghar Muft Bijli Yojana - solar rooftop

> This block integrates the 2024-2025 examinable demand and paper metadata. It is kept separate from the 2018-2023 block and does not convert an unkeyed/answer-free objective question into a solved answer.

Historical PYQ Integration (2018-2023)

> **Status:** Question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: `_PYQ-ROUTING-MAINS-GS1-GS2-ESSAY-2018-2023.md`, `_PYQ-ROUTING-MAINS-GS3-GS4-2018-2023.md`, `_PYQ-ROUTING-PRELIMS-2018-2023.md`. **Answer-key rule:** The official 2018-2023 Prelims/CSAT keys are not held locally; no option or answer has been inferred.

- **Years represented:** 2018, 2020, 2021, 2022, 2023
- **Paper(s):** GS-I, GS-III, Prelims GS-I
- **Routed question demands:** 11

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2018	Prelims GS-I	67	Solar power production silicon wafers and tariff regulation India	Objective question; official key unavailable locally	Cross-routed to solar-technology and energy-regulation owners; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.
2020	GS-I	16	Solar energy potential and its regional variations in India	Elaborate · 15 marks · 250 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
2020	GS-III	16	Solar energy benefits versus conventional energy and government initiatives	Describe · 15 marks · 250 words	Cross-routed to renewable-technology and exam-complete Core energy-infrastructure owner	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
2020	Prelims GS-I	84	Biofuels raw materials permitted under National Policy	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.
2020	Prelims GS-I	88	Solar water pumps surface submersible centrifugal piston types	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.
2021	GS-III	6	Green Grid Initiative purpose at COP26 and ISA origin	Explain · 10 marks · 150 words	Cross-routed to climate/solar and exam-complete Core grid-integration owners	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
2022	GS-I	7	Wind energy potential in India and its limited spatial spread	Examine and explain · 10 marks · 150 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
2022	GS-III	12	Renewable energy 2030 target and shift from fossil fuel subsidies	Justify · 15 marks · 250 words	Cross-routed to climate-transition and exam-complete Core subsidy/energy-market owner	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
2022	Prelims GS-I	84	Solar parks floating solar and airport projects Indian states	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.
2023	Prelims GS-I	60	Green hydrogen applications fuel blending and fuel cells	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.
2023	Prelims GS-I	99	Green hydrogen decarbonizing fertilizer oil refinery steel industries	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.

What this owner must now support

- Solar power production silicon wafers and tariff regulation India
- Solar energy potential and its regional variations in India
- Solar energy benefits versus conventional energy and government initiatives
- Biofuels raw materials permitted under National Policy

- Solar water pumps surface submersible centrifugal piston types
- Green Grid Initiative purpose at COP26 and ISA origin
- Wind energy potential in India and its limited spatial spread
- Renewable energy 2030 target and shift from fossil fuel subsidies
- Solar parks floating solar and airport projects Indian states
- Green hydrogen applications fuel blending and fuel cells
- Green hydrogen decarbonizing fertilizer oil refinery steel industries

> The table integrates the examinable demand and paper metadata. It does not turn an unkeyed objective question into a solved answer, and it does not claim that lexical presence alone proves full conceptual sufficiency.

10. PYQ-based analytical application

- ANALYSIS: Prelims questions on green/grey/blue hydrogen distinctions or National Green Hydrogen

Mission targets should be answered by applying the precise process/target definitions.

- ANALYSIS: Mains answers on "India's energy transition" should explicitly engage the intermittency/

storage challenge and the renewable-siting-versus-species-habitat tension to demonstrate analytical depth beyond describing capacity-addition targets.

Historical PYQ Integration (2018-2023)

> **Status:** Question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: `_PYQ-ROUTING-MAINS-GS1-GS2-ESSAY-2018-2023.md`, `_PYQ-ROUTING-MAINS-GS3-GS4-2018-2023.md`.

- **Years represented:** 2020, 2021, 2022
- **Paper(s):** GS-I, GS-III
- **Routed question demands:** 5

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2020	GS-I	16	Solar energy potential and its regional variations in India	Elaborate · 15 marks · 250 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
2020	GS-III	16	Solar energy benefits versus conventional energy and government initiatives	Describe · 15 marks · 250 words	Cross-routed to renewable-technology and exam-complete Core energy-infrastructure owner	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
2021	GS-III	6	Green Grid Initiative purpose at COP26 and ISA origin	Explain · 10 marks · 150 words	Cross-routed to climate/solar and exam-complete Core grid-integration owners	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
2022	GS-I	7	Wind energy potential in India and its limited spatial spread	Examine and explain · 10 marks · 150 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
2022	GS-III	12	Renewable energy 2030 target and shift from fossil fuel subsidies	Justify · 15 marks · 250 words	Cross-routed to climate-transition and exam-complete Core subsidy/energy-market owner	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.

What this owner must now support

- Solar energy potential and its regional variations in India
- Solar energy benefits versus conventional energy and government initiatives
- Green Grid Initiative purpose at COP26 and ISA origin
- Wind energy potential in India and its limited spatial spread
- Renewable energy 2030 target and shift from fossil fuel subsidies

> The table integrates the examinable demand and paper metadata. It does not turn an unkeyed objective question into a solved answer, and it does not claim that lexical presence alone proves full conceptual sufficiency.

PYQ DEMAND CARD 1 - 2025 GS-I

Demand: Explain ecological and economic benefits of solar energy generation in India.

Status: Verified routed Mains demand; no current capacity or generation value is inferred.

Model solution: Capacity-generation boundary: Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply. **Variable-firm distinction:** Solar and wind output varies with resource availability, whereas dispatchable or firm supply can be scheduled or supported to meet demand; nameplate capacity alone does not establish firmness. **Integration chain:** A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints. **Target-achievement boundary:** A gross announced target, sanctioned project pipeline, installed capacity, commissioned capacity and achieved generation are separate statuses that require separate dated evidence. **Current evidence boundary:** Installed capacity, generation, project cost, mission target, achieved output, electrolyser scale, hydrogen classification and certification status require a dated MNRE, power-system or applicable standards source. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

Demand decoding: The directive **answer** requires a direct position on "PYQ DEMAND CARD 1 - 2025 GS-I", every clause, exact ecological mechanism and scale, named Indian law or institution, dated treaty/policy status, evidence, trade-offs and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: Capacity-generation boundary: Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply. **Variable-firm distinction:** Solar and wind output varies with resource availability, whereas dispatchable or firm supply can be scheduled or supported to meet demand; nameplate capacity alone does not establish firmness. **Integration chain:** A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints. **Target-achievement boundary:** A gross announced target, sanctioned project pipeline, installed capacity, commissioned capacity and achieved generation are separate statuses that require separate dated evidence. **Current evidence boundary:** Installed capacity, generation, project cost, mission target, achieved output, electrolyser scale, hydrogen classification and certification status require a dated MNRE, power-system or applicable standards source. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

Analytical body:

1. **Claim and named evidence:** Demand: Explain ecological and economic benefits of solar energy generation in India. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
2. **Claim and named evidence:** Status: Verified routed Mains demand; no current capacity or generation value is inferred. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:**

State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.

Counter-position / limit: A designation, schedule, COP decision, policy target, budget, installed capacity, registration, treatment capacity or chronological association cannot alone establish ecological recovery, compliance, attribution or net climate benefit; test mechanism, monitoring, counterfactual and implementation.

Qualified conclusion: Capacity-generation boundary: Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply. **Variable-firm distinction:** Solar and wind output varies with resource availability, whereas dispatchable or firm supply can be scheduled or supported to meet demand; nameplate capacity alone does not establish firmness. **Integration chain:** A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints. **Target-achievement boundary:** A gross announced target, sanctioned project pipeline, installed capacity, commissioned capacity and achieved generation are separate statuses that require separate dated evidence. **Current evidence boundary:** Installed capacity, generation, project cost, mission target, achieved output, electrolyser scale, hydrogen classification and certification status require a dated MNRE, power-system or applicable standards source. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism/status -> institution/instrument -> implementation -> ecological and social outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for taxon, unit, baseline, date, jurisdiction, exception, causation and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains mechanisms rather than listing schemes, uses India-centric evidence and preserves ecological, species, legal, treaty, institutional, climate-unit, pollution-standard and causal distinctions.

How to improve this answer: For "PYQ DEMAND CARD 1 - 2025 GS-I", replace the weakest catalogue point with one exact mechanism or distinction, named law/institution/treaty/species, dated status, measurable outcome, implementation constraint and answer-specific qualification.

PYQ DEMAND CARD 2 - 2025 GS-III

Demand: Explain how clean technology can support energy independence and the role of biotechnology.

Status: Verified routed Mains demand; this topic owns the renewable-grid-hydrogen half only.

Model solution: Capacity-generation boundary: Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply. **Renewable-non-fossil boundary:** Solar, wind, hydro, biomass and geothermal are renewable sources, while non-fossil is a wider category that can include nuclear power; the two labels are not interchangeable. **Integration chain:** A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints. **Direct-electrification test:** Direct use of clean electricity is generally the first option where technically suitable, while hydrogen is considered where molecules, high-temperature heat, long-duration storage or weight constraints matter. **Hard-to-abate fit:** Fertiliser, refining, selected steel processes, shipping fuels and some heavy transport can use hydrogen or derivatives, but sector suitability and conversion losses must be assessed rather than assumed. **Mission-target-outcome boundary:** A mission target expresses intended future scale, while allocation, manufacturing award, commissioning, output and emissions outcome are separate evidence stages. **Current evidence boundary:** Installed capacity, generation, project cost, mission target, achieved output, electrolyser scale, hydrogen classification and certification status require a dated MNRE, power-system or applicable standards source. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

Demand decoding: The directive **answer** requires a direct position on "PYQ DEMAND CARD 2 - 2025 GS-III", every clause, exact ecological mechanism and scale, named Indian law or institution, dated treaty/policy status, evidence, trade-offs and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: Capacity-generation boundary: Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply. **Renewable-non-fossil boundary:** Solar, wind, hydro, biomass and geothermal are renewable sources, while non-fossil is a wider category that can include nuclear power; the two labels are not interchangeable. **Integration chain:** A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints. **Direct-electrification test:** Direct use of clean electricity is generally the first option where technically suitable, while hydrogen is considered where molecules, high-temperature heat, long-duration storage or weight constraints matter. **Hard-to-abate fit:** Fertiliser, refining, selected steel processes, shipping fuels and some heavy transport can use hydrogen or derivatives, but sector suitability and conversion losses must be assessed rather than assumed. **Mission-target-outcome boundary:** A mission target expresses intended future scale, while allocation, manufacturing award, commissioning, output and emissions outcome are separate evidence stages. **Current evidence boundary:** Installed capacity, generation, project cost, mission target, achieved output, electrolyser scale, hydrogen classification and certification status require a dated MNRE, power-system or applicable standards source. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

Analytical body:

1. **Claim and named evidence:** Demand: Explain how clean technology can support energy independence and the role of biotechnology. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
2. **Claim and named evidence:** Status: Verified routed Mains demand; this topic owns the renewable-grid-hydrogen half only. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.

Counter-position / limit: A designation, schedule, COP decision, policy target, budget, installed capacity, registration, treatment capacity or chronological association cannot alone establish ecological recovery, compliance, attribution or net climate benefit; test mechanism, monitoring, counterfactual and implementation.

Qualified conclusion: Capacity-generation boundary: Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply. **Renewable-non-fossil boundary:** Solar, wind, hydro, biomass and geothermal are renewable sources, while non-fossil is a wider category that can include nuclear power; the two labels are not interchangeable. **Integration chain:** A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints. **Direct-electrification test:** Direct use of clean electricity is generally the first option where technically suitable, while hydrogen is considered where molecules, high-temperature heat, long-duration storage or weight constraints matter. **Hard-to-abate fit:** Fertiliser, refining, selected steel processes, shipping fuels and some heavy transport can use hydrogen or derivatives, but sector suitability and conversion losses must be assessed rather than assumed. **Mission-target-outcome boundary:** A mission target expresses intended future scale, while allocation, manufacturing award, commissioning, output and emissions outcome are separate evidence stages. **Current evidence boundary:** Installed capacity, generation, project cost, mission target, achieved output, electrolyser scale, hydrogen classification and certification status require a dated MNRE, power-system or applicable standards source. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism/status -> institution/instrument -> implementation -> ecological and social outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for taxon, unit, baseline, date, jurisdiction, exception, causation and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains mechanisms rather than listing schemes, uses India-centric evidence and preserves ecological, species, legal, treaty, institutional, climate-unit, pollution-standard and causal distinctions.

How to improve this answer: For "PYQ DEMAND CARD 2 - 2025 GS-III", replace the weakest catalogue point with one exact mechanism or distinction, named law/institution/treaty/species, dated status, measurable outcome, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 1 - 10 MARKS

Question: Distinguish renewable installed capacity, generation and reliable supply. Answer in about 150 words.

Model thesis: Claim: Capacity-generation boundary. **Named evidence/example:** Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Variable-firm distinction. **Named evidence/example:** Solar and wind output varies with resource availability, whereas dispatchable or firm supply can be scheduled or supported to meet demand; nameplate capacity alone does not establish firmness. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Integration chain. **Named evidence/example:** A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Storage boundary. **Named evidence/example:** Battery and pumped-storage systems shift energy across time and provide grid services, but storage consumes previously generated energy and is not an independent primary energy source. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Claim -> named evidence -> analysis -> qualification:

- Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply.
- Solar and wind output varies with resource availability, whereas dispatchable or firm supply can be scheduled or supported to meet demand; nameplate capacity alone does not establish firmness.
- A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints.
- Battery and pumped-storage systems shift energy across time and provide grid services, but storage consumes previously generated energy and is not an independent primary energy source.

Qualified conclusion: Claim: Capacity-generation boundary. **Named evidence/example:** Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Variable-firm distinction. **Named evidence/example:** Solar and wind output varies with resource availability, whereas dispatchable or firm supply can be scheduled or supported to meet demand; nameplate capacity alone does not establish firmness. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Integration chain. **Named evidence/example:** A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Storage boundary. **Named evidence/example:** Battery and pumped-storage systems shift energy across time and provide grid services, but storage consumes previously generated energy and is not an independent primary energy source. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Demand decoding: The directive **answer** requires a direct position on "Distinguish renewable installed capacity, generation and reliable supply. Answer in about 150...", every clause, exact ecological mechanism and scale, named Indian law or institution, dated treaty/policy status, evidence, trade-offs and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Capacity-generation boundary. **Named evidence/example:** Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Variable-firm distinction. **Named evidence/example:** Solar and wind output varies with resource availability, whereas dispatchable or firm supply can be scheduled or supported to meet demand; nameplate capacity alone does not establish firmness. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Integration chain. **Named evidence/example:** A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Storage boundary. **Named evidence/example:** Battery and pumped-storage systems shift energy across time and provide grid services, but storage consumes previously generated energy and is not an independent primary energy source. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Analytical body:

1. **Claim and named evidence:** Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
2. **Claim and named evidence:** Solar and wind output varies with resource availability, whereas dispatchable or firm supply can be scheduled or supported to meet demand; nameplate capacity alone does not establish firmness. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:**

State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.

3. **Claim and named evidence:** A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
4. **Claim and named evidence:** Battery and pumped-storage systems shift energy across time and provide grid services, but storage consumes previously generated energy and is not an independent primary energy source. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.

Counter-position / limit: A designation, schedule, COP decision, policy target, budget, installed capacity, registration, treatment capacity or chronological association cannot alone establish ecological recovery, compliance, attribution or net climate benefit; test mechanism, monitoring, counterfactual and implementation.

Qualified conclusion: Claim: Capacity-generation boundary. **Named evidence/example:** Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Variable-firm distinction. **Named evidence/example:** Solar and wind output varies with resource availability, whereas dispatchable or firm supply can be scheduled or supported to meet demand; nameplate capacity alone does not establish firmness. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Integration chain. **Named evidence/example:** A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Storage boundary. **Named evidence/example:** Battery and pumped-storage systems shift energy across time and provide grid services, but storage consumes previously generated energy and is not an independent primary energy source. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Executable exam-length answer / compression plan: For 10 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism/status -> institution/instrument -> implementation -> ecological and social outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for taxon, unit, baseline, date, jurisdiction, exception, causation and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains mechanisms rather than listing schemes, uses India-centric evidence and preserves ecological, species, legal, treaty, institutional, climate-unit, pollution-standard and causal distinctions.

How to improve this answer: For "Distinguish renewable installed capacity, generation and reliable supply. Answer in about 150...", replace the weakest catalogue point with one exact mechanism or distinction, named law/institution/treaty/species, dated status, measurable outcome, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 2 - 10 MARKS

Question: Explain the electrolysis pathway and the evidence needed for a green-hydrogen claim. Answer in about 150 words.

Model thesis: Claim: Electrolysis pathway. **Named evidence/example:** Electrolysis uses electricity to split water into hydrogen and oxygen; calling the product green depends on the electricity and accounting pathway, not on the chemical identity of hydrogen. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Label-standard-certification boundary. **Named evidence/example:** A production pathway, a government definition or standard, a certification method and a market label are distinct layers; a pathway description alone does not prove certified compliance. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Hydrogen-colour boundary. **Named evidence/example:** Grey, blue and green are shorthand for different production and emissions-management pathways; colour language does not replace a stated lifecycle boundary, capture performance or official standard. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Electricity-attribute boundary. **Named evidence/example:** Claims about renewable electricity used for hydrogen may depend on sourcing, temporal matching, geography and additionality rules in the applicable standard; those attributes require documentary evidence. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Claim -> named evidence -> analysis -> qualification:

- Electrolysis uses electricity to split water into hydrogen and oxygen; calling the product green depends on the electricity and accounting pathway, not on the chemical identity of hydrogen.
- A production pathway, a government definition or standard, a certification method and a market label are distinct layers; a pathway description alone does not prove certified compliance.
- Grey, blue and green are shorthand for different production and emissions-management pathways; colour language does not replace a stated lifecycle boundary, capture performance or official standard.
- Claims about renewable electricity used for hydrogen may depend on sourcing, temporal matching, geography and additionality rules in the applicable standard; those attributes require documentary evidence.

Qualified conclusion: Claim: Electrolysis pathway. **Named evidence/example:** Electrolysis uses electricity to split water into hydrogen and oxygen; calling the product green depends on the electricity and accounting pathway, not on the chemical identity of hydrogen. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Label-standard-certification boundary. **Named evidence/example:** A production pathway, a government definition or standard, a certification method and a market label are distinct layers; a pathway description alone does not prove certified compliance. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Hydrogen-colour boundary. **Named evidence/example:** Grey, blue and green are shorthand for different production and emissions-management pathways; colour language does not replace a stated lifecycle boundary, capture performance or official standard. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Electricity-attribute boundary. **Named**

evidence/example: Claims about renewable electricity used for hydrogen may depend on sourcing, temporal matching, geography and additionality rules in the applicable standard; those attributes require documentary evidence. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Demand decoding: The directive **explain** requires a direct position on "Explain the electrolysis pathway and the evidence needed for a green-hydrogen claim. Answer...", every clause, exact ecological mechanism and scale, named Indian law or institution, dated treaty/policy status, evidence, trade-offs and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Electrolysis pathway. **Named evidence/example:** Electrolysis uses electricity to split water into hydrogen and oxygen; calling the product green depends on the electricity and accounting pathway, not on the chemical identity of hydrogen. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Label-standard-certification boundary. **Named evidence/example:** A production pathway, a government definition or standard, a certification method and a market label are distinct layers; a pathway description alone does not prove certified compliance. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Hydrogen-colour boundary. **Named evidence/example:** Grey, blue and green are shorthand for different production and emissions-management pathways; colour language does not replace a stated lifecycle boundary, capture performance or official standard. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Electricity-attribute boundary. **Named evidence/example:** Claims about renewable electricity used for hydrogen may depend on sourcing, temporal matching, geography and additionality rules in the applicable standard; those attributes require documentary evidence. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Analytical body:

1. **Claim and named evidence:** Electrolysis uses electricity to split water into hydrogen and oxygen; calling the product green depends on the electricity and accounting pathway, not on the chemical identity of hydrogen. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
2. **Claim and named evidence:** A production pathway, a government definition or standard, a certification method and a market label are distinct layers; a pathway description alone does not prove certified compliance. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
3. **Claim and named evidence:** Grey, blue and green are shorthand for different production and emissions-management pathways; colour language does not replace a stated lifecycle boundary, capture performance or official standard. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.

4. **Claim and named evidence:** Claims about renewable electricity used for hydrogen may depend on sourcing, temporal matching, geography and additionality rules in the applicable standard; those attributes require documentary evidence. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.

Counter-position / limit: A designation, schedule, COP decision, policy target, budget, installed capacity, registration, treatment capacity or chronological association cannot alone establish ecological recovery, compliance, attribution or net climate benefit; test mechanism, monitoring, counterfactual and implementation.

Qualified conclusion: Claim: Electrolysis pathway. **Named evidence/example:** Electrolysis uses electricity to split water into hydrogen and oxygen; calling the product green depends on the electricity and accounting pathway, not on the chemical identity of hydrogen. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Label-standard-certification boundary. **Named evidence/example:** A production pathway, a government definition or standard, a certification method and a market label are distinct layers; a pathway description alone does not prove certified compliance. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Hydrogen-colour boundary. **Named evidence/example:** Grey, blue and green are shorthand for different production and emissions-management pathways; colour language does not replace a stated lifecycle boundary, capture performance or official standard. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Electricity-attribute boundary. **Named evidence/example:** Claims about renewable electricity used for hydrogen may depend on sourcing, temporal matching, geography and additionality rules in the applicable standard; those attributes require documentary evidence. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Executable exam-length answer / compression plan: For 10 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism/status -> institution/instrument -> implementation -> ecological and social outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for taxon, unit, baseline, date, jurisdiction, exception, causation and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains mechanisms rather than listing schemes, uses India-centric evidence and preserves ecological, species, legal, treaty, institutional, climate-unit, pollution-standard and causal distinctions.

How to improve this answer: For "Explain the electrolysis pathway and the evidence needed for a green-hydrogen claim. Answer...", replace the weakest catalogue point with one exact mechanism or distinction, named law/institution/treaty/species, dated status, measurable outcome, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 3 - 15 MARKS

Question: Assess grid integration requirements for variable renewable energy. Answer in about 250 words.

Model thesis: Claim: Capacity-generation boundary. **Named evidence/example:** Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Variable-firm distinction. **Named evidence/example:** Solar and wind output varies with resource availability, whereas dispatchable or firm supply can be scheduled or supported to meet demand; nameplate capacity alone does not establish firmness. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Integration chain. **Named evidence/example:** A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Storage boundary. **Named evidence/example:** Battery and pumped-storage systems shift energy across time and provide grid services, but storage consumes previously generated energy and is not an independent primary energy source. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Target-achievement boundary. **Named evidence/example:** A gross announced target, sanctioned project pipeline, installed capacity, commissioned capacity and achieved generation are separate statuses that require separate dated evidence. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Claim -> named evidence -> analysis -> qualification:

- Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply.
- Solar and wind output varies with resource availability, whereas dispatchable or firm supply can be scheduled or supported to meet demand; nameplate capacity alone does not establish firmness.
- A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints.
- Battery and pumped-storage systems shift energy across time and provide grid services, but storage consumes previously generated energy and is not an independent primary energy source.
- A gross announced target, sanctioned project pipeline, installed capacity, commissioned capacity and achieved generation are separate statuses that require separate dated evidence.

Qualified conclusion: Claim: Capacity-generation boundary. **Named evidence/example:** Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Variable-firm distinction. **Named evidence/example:** Solar and wind output varies with resource availability, whereas dispatchable or firm supply can be scheduled or supported to meet demand; nameplate capacity alone does not establish firmness. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Integration chain. **Named evidence/example:** A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Storage boundary. **Named evidence/example:** Battery and pumped-storage systems shift energy across time and provide grid services, but storage consumes previously generated energy and is not an independent primary energy source. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Target-achievement boundary. **Named evidence/example:** A gross announced target, sanctioned project pipeline, installed capacity, commissioned capacity and achieved generation are separate statuses that require separate dated evidence. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Demand decoding: The directive **assess** requires a direct position on "Assess grid integration requirements for variable renewable energy. Answer in about 250 words.", every clause, exact ecological mechanism and scale, named Indian law or institution, dated treaty/policy status, evidence, trade-offs and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Capacity-generation boundary. **Named evidence/example:** Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Variable-firm distinction. **Named evidence/example:** Solar and wind output varies with resource availability, whereas dispatchable or firm supply can be scheduled or supported to meet demand; nameplate capacity alone does not establish firmness. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Integration chain. **Named evidence/example:** A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Storage boundary. **Named evidence/example:** Battery and pumped-storage systems shift energy across time and provide grid services, but storage consumes previously generated energy and is not an independent primary energy source. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Target-achievement boundary. **Named evidence/example:** A gross announced target, sanctioned project pipeline, installed capacity, commissioned capacity and achieved generation are separate statuses that require separate dated evidence. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Analytical body:

1. **Claim and named evidence:** Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
2. **Claim and named evidence:** Solar and wind output varies with resource availability, whereas dispatchable or firm supply can be scheduled or supported to meet demand; nameplate capacity alone does not establish firmness. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
3. **Claim and named evidence:** A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
4. **Claim and named evidence:** Battery and pumped-storage systems shift energy across time and provide grid services, but storage consumes previously generated energy and is not an independent primary energy source. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
5. **Claim and named evidence:** A gross announced target, sanctioned project pipeline, installed capacity, commissioned capacity and achieved generation are separate statuses that require separate dated evidence. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.

Counter-position / limit: A designation, schedule, COP decision, policy target, budget, installed capacity, registration, treatment capacity or chronological association cannot alone establish ecological recovery, compliance, attribution or net climate benefit; test mechanism, monitoring, counterfactual and implementation.

Qualified conclusion: Claim: Capacity-generation boundary. **Named evidence/example:** Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Variable-firm distinction. **Named evidence/example:** Solar and wind output varies with resource availability, whereas dispatchable or firm supply can be scheduled or supported to meet demand; nameplate capacity alone does not establish firmness. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Integration chain. **Named evidence/example:** A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Storage boundary. **Named evidence/example:** Battery and pumped-storage systems shift energy across time and provide grid services, but storage consumes previously generated energy and is not an independent primary energy source. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Target-achievement boundary. **Named evidence/example:** A gross announced target, sanctioned project pipeline, installed capacity, commissioned capacity and achieved generation are separate statuses that require separate dated evidence. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism/status -> institution/instrument -> implementation -> ecological and social outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for taxon, unit, baseline, date, jurisdiction, exception, causation and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains mechanisms rather than listing schemes, uses India-centric evidence and preserves ecological, species, legal, treaty, institutional, climate-unit, pollution-standard and causal distinctions.

How to improve this answer: For "Assess grid integration requirements for variable renewable energy. Answer in about 250 words.", replace the weakest catalogue point with one exact mechanism or distinction, named law/institution/treaty/species, dated status, measurable outcome, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 4 - 15 MARKS

Question: Explain why hydrogen should be targeted selectively at hard-to-abate uses. Answer in about 250 words.

Model thesis: Claim: Direct-electrification test. **Named evidence/example:** Direct use of clean electricity is generally the first option where technically suitable, while hydrogen is considered where molecules, high-temperature heat, long-duration storage or weight constraints matter. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Hard-to-abate fit. **Named evidence/example:** Fertiliser, refining, selected steel processes, shipping fuels and some heavy transport can use hydrogen or derivatives, but sector suitability and conversion losses must be assessed rather than assumed. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Derivative boundary. **Named evidence/example:** Hydrogen, ammonia, methanol and synthetic fuels are related but different products with different conversion steps, handling requirements and end uses; a hydrogen target is not automatically a derivative-output target. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Safety-and-logistics boundary. **Named evidence/example:** Hydrogen production, compression, liquefaction, storage, pipelines, ports and end use create distinct engineering and safety requirements; production capacity does not prove delivered usable fuel. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Claim -> named evidence -> analysis -> qualification:

- Direct use of clean electricity is generally the first option where technically suitable, while hydrogen is considered where molecules, high-temperature heat, long-duration storage or weight constraints matter.
- Fertiliser, refining, selected steel processes, shipping fuels and some heavy transport can use hydrogen or derivatives, but sector suitability and conversion losses must be assessed rather than assumed.
- Hydrogen, ammonia, methanol and synthetic fuels are related but different products with different conversion steps, handling requirements and end uses; a hydrogen target is not automatically a derivative-output target.

- Hydrogen production, compression, liquefaction, storage, pipelines, ports and end use create distinct engineering and safety requirements; production capacity does not prove delivered usable fuel.

Qualified conclusion: Claim: Direct-electrification test. **Named evidence/example:** Direct use of clean electricity is generally the first option where technically suitable, while hydrogen is considered where molecules, high-temperature heat, long-duration storage or weight constraints matter. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Hard-to-abate fit. **Named evidence/example:** Fertiliser, refining, selected steel processes, shipping fuels and some heavy transport can use hydrogen or derivatives, but sector suitability and conversion losses must be assessed rather than assumed. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Derivative boundary. **Named evidence/example:** Hydrogen, ammonia, methanol and synthetic fuels are related but different products with different conversion steps, handling requirements and end uses; a hydrogen target is not automatically a derivative-output target. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Safety-and-logistics boundary. **Named evidence/example:** Hydrogen production, compression, liquefaction, storage, pipelines, ports and end use create distinct engineering and safety requirements; production capacity does not prove delivered usable fuel. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Demand decoding: The directive **explain** requires a direct position on "Explain why hydrogen should be targeted selectively at hard-to-abate uses. Answer in about...", every clause, exact ecological mechanism and scale, named Indian law or institution, dated treaty/policy status, evidence, trade-offs and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Direct-electrification test. **Named evidence/example:** Direct use of clean electricity is generally the first option where technically suitable, while hydrogen is considered where molecules, high-temperature heat, long-duration storage or weight constraints matter. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Hard-to-abate fit. **Named evidence/example:** Fertiliser, refining, selected steel processes, shipping fuels and some heavy transport can use hydrogen or derivatives, but sector suitability and conversion losses must be assessed rather than assumed. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Derivative boundary. **Named evidence/example:** Hydrogen, ammonia, methanol and synthetic fuels are related but different products with different conversion steps, handling requirements and end uses; a hydrogen target is not automatically a derivative-output target. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Safety-and-logistics boundary. **Named evidence/example:** Hydrogen production, compression, liquefaction, storage, pipelines, ports and end use create distinct engineering and safety requirements; production capacity does not prove delivered usable fuel. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Analytical body:

1. **Claim and named evidence:** Direct use of clean electricity is generally the first option where technically suitable, while hydrogen is considered where molecules, high-temperature heat, long-duration storage or weight constraints matter. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
2. **Claim and named evidence:** Fertiliser, refining, selected steel processes, shipping fuels and some heavy transport can use hydrogen or derivatives, but sector suitability and conversion losses must be assessed rather than assumed. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
3. **Claim and named evidence:** Hydrogen, ammonia, methanol and synthetic fuels are related but different products with different conversion steps, handling requirements and end uses; a hydrogen target is not automatically a derivative-output target. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
4. **Claim and named evidence:** Hydrogen production, compression, liquefaction, storage, pipelines, ports and end use create distinct engineering and safety requirements; production capacity does not prove delivered usable fuel. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.

Counter-position / limit: A designation, schedule, COP decision, policy target, budget, installed capacity, registration, treatment capacity or chronological association cannot alone establish ecological recovery, compliance, attribution or net climate benefit; test mechanism, monitoring, counterfactual and implementation.

Qualified conclusion: Claim: Direct-electrification test. **Named evidence/example:** Direct use of clean electricity is generally the first option where technically suitable, while hydrogen is considered where molecules, high-temperature heat, long-duration storage or weight constraints matter. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Hard-to-abate fit. **Named evidence/example:** Fertiliser, refining, selected steel processes, shipping fuels and some heavy transport can use hydrogen or derivatives, but sector suitability and conversion losses must be assessed rather than assumed. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Derivative boundary. **Named evidence/example:** Hydrogen, ammonia, methanol and synthetic fuels are related but different products with different conversion steps, handling requirements and end uses; a hydrogen target is not automatically a derivative-output target. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Safety-and-logistics boundary. **Named evidence/example:** Hydrogen production, compression, liquefaction, storage, pipelines, ports and end use create distinct engineering and safety requirements; production capacity does not prove delivered usable fuel. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism/status -> institution/instrument -> implementation -> ecological and social outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for taxon, unit, baseline, date, jurisdiction, exception, causation and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains mechanisms rather than listing schemes, uses India-centric evidence and preserves ecological, species, legal, treaty, institutional, climate-unit, pollution-standard and causal distinctions.

How to improve this answer: For "Explain why hydrogen should be targeted selectively at hard-to-abate uses. Answer in about...", replace the weakest catalogue point with one exact mechanism or distinction, named law/institution/treaty/species, dated status, measurable outcome, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 5 - 20 MARKS

Question: Evaluate the National Green Hydrogen Mission through target-to-outcome discipline. Answer in about 300 words.

Model thesis: Claim: Target-achievement boundary. **Named evidence/example:** A gross announced target, sanctioned project pipeline, installed capacity, commissioned capacity and achieved generation are separate statuses that require separate dated evidence. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Status-verb discipline. **Named evidence/example:** Announced, approved, allocated, awarded, under construction, commissioned and producing describe different project stages; one stage must never be substituted for another. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Label-standard-certification boundary. **Named evidence/example:** A production pathway, a government definition or standard, a certification method and a market label are distinct layers; a pathway description alone does not prove certified compliance. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Mission-target-outcome boundary. **Named evidence/example:** A mission target expresses intended future scale, while allocation, manufacturing award, commissioning, output and emissions outcome are separate evidence stages. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** SIGHT-dual-pillar boundary. **Named evidence/example:** The SIGHT programme distinguishes support for electrolyser manufacturing from support for green-hydrogen production; manufacturing capability and hydrogen output are related but not identical. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Current evidence boundary. **Named evidence/example:** Installed capacity, generation, project cost, mission target, achieved output, electrolyser scale, hydrogen classification and certification status require a dated MNRE, power-system or applicable standards source. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Claim -> named evidence -> analysis -> qualification:

- A gross announced target, sanctioned project pipeline, installed capacity, commissioned capacity and achieved generation are separate statuses that require separate dated evidence.
- Announced, approved, allocated, awarded, under construction, commissioned and producing describe different project stages; one stage must never be substituted for another.

- A production pathway, a government definition or standard, a certification method and a market label are distinct layers; a pathway description alone does not prove certified compliance.
- A mission target expresses intended future scale, while allocation, manufacturing award, commissioning, output and emissions outcome are separate evidence stages.
- The SIGHT programme distinguishes support for electrolyser manufacturing from support for green-hydrogen production; manufacturing capability and hydrogen output are related but not identical.
- Installed capacity, generation, project cost, mission target, achieved output, electrolyser scale, hydrogen classification and certification status require a dated MNRE, power-system or applicable standards source.

Qualified conclusion: Claim: Target-achievement boundary. **Named evidence/example:** A gross announced target, sanctioned project pipeline, installed capacity, commissioned capacity and achieved generation are separate statuses that require separate dated evidence. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Status-verb discipline. **Named evidence/example:** Announced, approved, allocated, awarded, under construction, commissioned and producing describe different project stages; one stage must never be substituted for another. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Label-standard-certification boundary. **Named evidence/example:** A production pathway, a government definition or standard, a certification method and a market label are distinct layers; a pathway description alone does not prove certified compliance. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Mission-target-outcome boundary. **Named evidence/example:** A mission target expresses intended future scale, while allocation, manufacturing award, commissioning, output and emissions outcome are separate evidence stages. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** SIGHT-dual-pillar boundary. **Named evidence/example:** The SIGHT programme distinguishes support for electrolyser manufacturing from support for green-hydrogen production; manufacturing capability and hydrogen output are related but not identical. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Current evidence boundary. **Named evidence/example:** Installed capacity, generation, project cost, mission target, achieved output, electrolyser scale, hydrogen classification and certification status require a dated MNRE, power-system or applicable standards source. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Demand decoding: The directive **evaluate** requires a direct position on "Evaluate the National Green Hydrogen Mission through target-to-outcome discipline. Answer in...", every clause, exact ecological mechanism and scale, named Indian law or institution, dated treaty/policy status, evidence, trade-offs and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Target-achievement boundary. **Named evidence/example:** A gross announced target, sanctioned project pipeline, installed capacity, commissioned capacity and achieved generation are separate statuses that require separate dated evidence. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Status-verb discipline. **Named evidence/example:** Announced, approved, allocated, awarded, under construction, commissioned and producing describe different project stages; one stage must never be substituted for another. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Label-standard-certification boundary. **Named evidence/example:** A production pathway, a government definition or standard, a certification method and a market label are distinct layers; a pathway description alone does not prove certified compliance. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Mission-target-outcome boundary. **Named evidence/example:** A mission target expresses intended future scale, while allocation, manufacturing award, commissioning, output and emissions outcome are separate evidence stages. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** SIGHT-dual-pillar boundary. **Named evidence/example:** The SIGHT programme distinguishes support for electrolyser manufacturing from support for green-hydrogen production; manufacturing capability and hydrogen output are related but not identical. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Current evidence boundary. **Named evidence/example:** Installed capacity, generation, project cost, mission target, achieved output, electrolyser scale, hydrogen classification and certification status require a dated MNRE, power-system or applicable standards source. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Analytical body:

1. **Claim and named evidence:** A gross announced target, sanctioned project pipeline, installed capacity, commissioned capacity and achieved generation are separate statuses that require separate dated evidence. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
2. **Claim and named evidence:** Announced, approved, allocated, awarded, under construction, commissioned and producing describe different project stages; one stage must never be substituted for another. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
3. **Claim and named evidence:** A production pathway, a government definition or standard, a certification method and a market label are distinct layers; a pathway description alone does not prove certified compliance. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
4. **Claim and named evidence:** A mission target expresses intended future scale, while allocation, manufacturing award, commissioning, output and emissions outcome are separate evidence stages. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
5. **Claim and named evidence:** The SIGHT programme distinguishes support for electrolyser manufacturing from support for green-hydrogen production; manufacturing capability and hydrogen output are related but not identical. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.

6. **Claim and named evidence:** Installed capacity, generation, project cost, mission target, achieved output, electrolyser scale, hydrogen classification and certification status require a dated MNRE, power-system or applicable standards source. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.

Counter-position / limit: A designation, schedule, COP decision, policy target, budget, installed capacity, registration, treatment capacity or chronological association cannot alone establish ecological recovery, compliance, attribution or net climate benefit; test mechanism, monitoring, counterfactual and implementation.

Qualified conclusion: Claim: Target-achievement boundary. **Named evidence/example:** A gross announced target, sanctioned project pipeline, installed capacity, commissioned capacity and achieved generation are separate statuses that require separate dated evidence. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Status-verb discipline. **Named evidence/example:** Announced, approved, allocated, awarded, under construction, commissioned and producing describe different project stages; one stage must never be substituted for another. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Label-standard-certification boundary. **Named evidence/example:** A production pathway, a government definition or standard, a certification method and a market label are distinct layers; a pathway description alone does not prove certified compliance. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Mission-target-outcome boundary. **Named evidence/example:** A mission target expresses intended future scale, while allocation, manufacturing award, commissioning, output and emissions outcome are separate evidence stages. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** SIGHT-dual-pillar boundary. **Named evidence/example:** The SIGHT programme distinguishes support for electrolyser manufacturing from support for green-hydrogen production; manufacturing capability and hydrogen output are related but not identical. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Current evidence boundary. **Named evidence/example:** Installed capacity, generation, project cost, mission target, achieved output, electrolyser scale, hydrogen classification and certification status require a dated MNRE, power-system or applicable standards source. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Executable exam-length answer / compression plan: For 20 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism/status -> institution/instrument -> implementation -> ecological and social outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for taxon, unit, baseline, date, jurisdiction, exception, causation and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains mechanisms rather than listing schemes, uses India-centric evidence and preserves ecological, species, legal, treaty, institutional, climate-unit, pollution-standard and causal distinctions.

How to improve this answer: For "Evaluate the National Green Hydrogen Mission through target-to-outcome discipline. Answer in...", replace the weakest catalogue point with one exact mechanism or distinction, named law/institution/treaty/species, dated status, measurable outcome, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 6 - 20 MARKS

Question: Design a sequenced renewable-electricity-to-green-hydrogen transition strategy. Answer in about 300 words.

Model thesis: Claim: Capacity-generation boundary. **Named evidence/example:** Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Renewable-non-fossil boundary. **Named evidence/example:** Solar, wind, hydro, biomass and geothermal are renewable sources, while non-fossil is a wider category that can include nuclear power; the two labels are not interchangeable. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Source-carrier boundary. **Named evidence/example:** Renewable electricity is an energy source output, while hydrogen is an energy carrier produced using an input pathway; hydrogen is not a primary renewable resource by itself. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Integration chain. **Named evidence/example:** A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Electrolysis pathway. **Named evidence/example:** Electrolysis uses electricity to split water into hydrogen and oxygen; calling the product green depends on the electricity and accounting pathway, not on the chemical identity of hydrogen. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Direct-electrification test. **Named evidence/example:** Direct use of clean electricity is generally the first option where technically suitable, while hydrogen is considered where molecules, high-temperature heat, long-duration storage or weight constraints matter. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Hard-to-abate fit. **Named evidence/example:** Fertiliser, refining, selected steel processes, shipping fuels and some heavy transport can use hydrogen or derivatives, but sector suitability and conversion losses must be assessed rather than assumed. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Water-context boundary. **Named evidence/example:** Electrolysis requires water and upstream treatment, but project-level water stress depends on source, quality, location, competing demand and management; no universal water-impact value should be inferred. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Safety-and-logistics boundary. **Named evidence/example:** Hydrogen production, compression, liquefaction, storage, pipelines, ports and end use create distinct engineering and safety requirements; production capacity does not prove delivered usable fuel. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Current evidence boundary. **Named evidence/example:** Installed capacity, generation, project cost, mission target, achieved output, electrolyser scale, hydrogen classification and certification status require a dated MNRE, power-system or applicable standards source. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Claim -> named evidence -> analysis -> qualification:

- Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply.
- Solar, wind, hydro, biomass and geothermal are renewable sources, while non-fossil is a wider category that can include nuclear power; the two labels are not interchangeable.
- Renewable electricity is an energy source output, while hydrogen is an energy carrier produced using an input pathway; hydrogen is not a primary renewable resource by itself.
- A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints.
- Electrolysis uses electricity to split water into hydrogen and oxygen; calling the product green depends on the electricity and accounting pathway, not on the chemical identity of hydrogen.
- Direct use of clean electricity is generally the first option where technically suitable, while hydrogen is considered where molecules, high-temperature heat, long-duration storage or weight constraints matter.
- Fertiliser, refining, selected steel processes, shipping fuels and some heavy transport can use hydrogen or derivatives, but sector suitability and conversion losses must be assessed rather than assumed.
- Electrolysis requires water and upstream treatment, but project-level water stress depends on source, quality, location, competing demand and management; no universal water-impact value should be inferred.
- Hydrogen production, compression, liquefaction, storage, pipelines, ports and end use create distinct engineering and safety requirements; production capacity does not prove delivered usable fuel.
- Installed capacity, generation, project cost, mission target, achieved output, electrolyser scale, hydrogen classification and certification status require a dated MNRE, power-system or applicable standards source.

Qualified conclusion: Claim: Capacity-generation boundary. **Named evidence/example:** Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Renewable-non-fossil boundary. **Named evidence/example:** Solar, wind, hydro, biomass and geothermal are renewable sources, while non-fossil is a wider category that can include nuclear power; the two labels are not interchangeable. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Source-carrier boundary. **Named evidence/example:** Renewable electricity is an energy source output, while hydrogen is an energy carrier produced using an input pathway; hydrogen is not a primary renewable resource by itself. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Integration chain. **Named evidence/example:** A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Electrolysis pathway. **Named evidence/example:** Electrolysis uses electricity to split water into hydrogen and oxygen; calling the product green depends on the electricity and accounting pathway, not on the chemical identity of hydrogen. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Direct-electrification test. **Named evidence/example:** Direct use of clean electricity is generally the first option where technically suitable, while hydrogen is considered where molecules, high-temperature heat, long-duration storage or weight constraints matter. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Hard-to-abate fit. **Named evidence/example:** Fertiliser, refining, selected steel processes, shipping fuels and some heavy transport can use hydrogen or derivatives, but sector suitability and conversion losses must be assessed rather than assumed. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Water-context boundary. **Named evidence/example:** Electrolysis requires water and upstream treatment, but project-level water stress depends on source, quality, location, competing demand and management; no universal water-impact value should be inferred. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Safety-and-logistics boundary. **Named evidence/example:** Hydrogen production, compression, liquefaction, storage, pipelines, ports and end use create distinct engineering and safety requirements; production capacity does not prove delivered usable fuel. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Current evidence boundary. **Named evidence/example:** Installed capacity, generation, project cost, mission target, achieved output, electrolyser scale, hydrogen classification and certification status require a dated MNRE, power-system or applicable standards source. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Demand decoding: The directive **answer** requires a direct position on "Design a sequenced renewable-electricity-to-green-hydrogen transition strategy. Answer in...", every clause, exact ecological mechanism and scale, named Indian law or institution, dated treaty/policy status, evidence, trade-offs and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Capacity-generation boundary. **Named evidence/example:** Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Renewable-non-fossil boundary. **Named evidence/example:** Solar, wind, hydro, biomass and geothermal are renewable sources, while non-fossil is a wider category that can include nuclear power; the two labels are not interchangeable. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Source-carrier boundary. **Named evidence/example:** Renewable electricity is an energy source output, while hydrogen is an energy carrier produced using an input pathway; hydrogen is not a primary renewable resource by itself. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Integration chain. **Named evidence/example:** A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Electrolysis pathway. **Named evidence/example:** Electrolysis uses electricity to split water into hydrogen and oxygen; 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no universal water-impact value should be inferred. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Safety-and-logistics boundary. **Named evidence/example:** Hydrogen production, compression, liquefaction, storage, pipelines, ports and end use create distinct engineering and safety requirements; production capacity does not prove delivered usable fuel. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Current evidence boundary. **Named evidence/example:** Installed capacity, generation, project cost, mission target, achieved output, electrolyser scale, hydrogen classification and certification status require a dated MNRE, power-system or applicable standards source. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Analytical body:

1. **Claim and named evidence:** Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
2. **Claim and named evidence:** Solar, wind, hydro, biomass and geothermal are renewable sources, while non-fossil is a wider category that can include nuclear power; the two labels are not interchangeable. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.

3. **Claim and named evidence:** Renewable electricity is an energy source output, while hydrogen is an energy carrier produced using an input pathway; hydrogen is not a primary renewable resource by itself. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
4. **Claim and named evidence:** A renewable-heavy power system links generation, forecasting, transmission, flexible demand, balancing resources and storage; adding generation without the rest can leave congestion or reliability constraints. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
5. **Claim and named evidence:** Electrolysis uses electricity to split water into hydrogen and oxygen; calling the product green depends on the electricity and accounting pathway, not on the chemical identity of hydrogen. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
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9. **Claim and named evidence:** Hydrogen production, compression, liquefaction, storage, pipelines, ports and end use create distinct engineering and safety requirements; production capacity does not prove delivered usable fuel. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
10. **Claim and named evidence:** Installed capacity, generation, project cost, mission target, achieved output, electrolyser scale, hydrogen classification and certification status require a dated MNRE, power-system or applicable standards source. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.

Counter-position / limit: A designation, schedule, COP decision, policy target, budget, installed capacity, registration, treatment capacity or chronological association cannot alone establish ecological recovery, compliance, attribution or net climate benefit; test mechanism, monitoring, counterfactual and implementation.

Qualified conclusion: Claim: Capacity-generation boundary. **Named evidence/example:** Installed capacity is nameplate power capability, while generation is electrical energy actually produced over time; neither capacity share nor capacity addition proves generation share or reliable supply. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Renewable-non-fossil boundary. **Named evidence/example:** Solar, wind, hydro, biomass and geothermal are renewable sources, while non-fossil is a wider category that can include nuclear power; the two labels are not interchangeable. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Source-carrier boundary. **Named evidence/example:** Renewable electricity is an energy source output, while hydrogen is an energy carrier produced using an input pathway; 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Executable exam-length answer / compression plan: For 20 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism/status -> institution/instrument -> implementation -> ecological and social outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for taxon, unit, baseline, date, jurisdiction, exception, causation and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains mechanisms rather than listing schemes, uses India-centric evidence and preserves ecological, species, legal, treaty, institutional, climate-unit, pollution-standard and causal distinctions.

How to improve this answer: For "Design a sequenced renewable-electricity-to-green-hydrogen transition strategy. Answer in...", replace the weakest catalogue point with one exact mechanism or distinction, named law/institution/treaty/species, dated status, measurable outcome, implementation constraint and answer-specific qualification.